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Automated mechanical cardiopulmonary resuscitation devices versus manual chest compressions in the treatment of cardiac arrest: Protocol of a systematic review and meta-analysis comparing machine to man

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Automated mechanical cardiopulmonary resuscitation devices versus manual chest compressions in the treatment of cardiac arrest:

Protocol of a systematic review and meta-analysis comparing machine to man

TITLE PAGE

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ABSTRACT

Introduction:

Cardiac arrest is a leading cause of death in industrialized countries. Cardiopulmonary Resuscitation (CPR) guidelines follow the principles of closed chest compression as described in 1960. Mechanical CPR devices are designed to improve chest compression quality, thus considered to improve resuscitation outcomes.

This protocol outlines a systematic review and meta-analysis methodology to assess trials investigating the therapeutic effect of automated mechanical CPR devices on the rate of Return of Spontaneous Circulation (ROSC), neurological state, and secondary endpoints (including short- and long-term survival rates, injuries through chest compression, and surrogate parameters for CPR performance and quality), compared to manual chest compressions in adults with cardiac arrest.

Methods and Analysis:

A sensitive search strategy is employed in established bibliographic databases, followed by forward and backward reference searching. Into qualitative analysis, we include randomized and quasi-randomized trials comparing mechanical to manual CPR. Into qualitative analysis, studies reporting any primary endpoint are included. Case series, case control-studies, and longitudinal cohort studies are excluded from meta-analysis.

Two reviwers independently assess publications using a pre-defined data collection form according to Cochrane recommendations. Standardized tools are used for data extraction, risks of bias are assessed using the ROBINS-I tool, and quality of evidence using the GRADE tool. If enough studies are identified for meta-analysis, the measures of association are calculated by dint of bivariate random-effects models. Statistical heterogeneity is evaluated

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with I^2 statistics and explored through sensitivity analysis. Reporting follows the PRISMA statement.

Ethics and dissemination:

No ethical approval will be needed because data from previous studies in will be retrieved and analysed. Most resuscitation studies are conducted under an emergency exception for informed consent. The publication contains data deriving from a dissertation project at Ulm University. We will disseminate the results through publication in a peer-reviewed journal and scientific conferences.

Registration:

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ARTICLE SUMMARY

Strengths and limitations of this study

- Numerous studies of heterogeneous design and quality imply the need for this systematic review and meta-analysis.
- Exensive and differentiated multi-step search strategy including 12 databases.
- Beside mechanical and manual chest compression, mulitpile confounding factors have influence on the outcome.
- Potential clarification of which patients under which conditions profit from which resuscitation technique may support threatment decisions.

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INTRODUCTION

For thousands of years, mankind has struggled to "re-animate" people showing no signs of life. Some of the oldest attempts documented can be found in the Bible, works from Galen of Pergamon, and Hippocrates of Cos. [1] In the evolution of Advanced Life Support (ALS), some procedures could convince: Among them are artificial respiration, closed-chest cardiac massage, and defibrillation, for example. [2 3] They found their way into the International Consensus on Cardiopulmonary Resuscitation (CPR) and Emergency Cardiovascular Care Science with Treatment Recommendations (CoSTR). [4] Other approaches had been discontinued, like "fumigation" for example: Although this concept was very obvious, to apply an "animating" substance into the "centre of life", the rectal application of tobacco smoke into the large intestine could not prevail with reference to patient outcome. [1]

A more recent development in emergency medicine is mechanical CPR: Devices that perform chest compressions mechanically and automatically through inflatable vests, pistons, or load distributing bands. Will mechanical CPR devices find their way into ALS algorithms and should ambulances, emergency departments, and critical care units routinely be equipped with these devices? Or are they just another approach whose fate would soon be sealed?

Rationale

Death following chronic coronary heart disease (CHD) and death following acute manifestation of CHD (that is acute myocardial infarction, presenting as sudden cardiac arrest) are the two most common causes of death in Germany. [5] Prevalence of sudden cardiac arrest is 80-150 per 100,000 person years. Put more clearly, in Germany, 70,000-100,000 patients suffer sudden cardiac arrest every year. Mortality rates are as high as 80-90%. [5-7]

If contemplated more specific, incidence of CHD is up to 600 per 100,000 persons and year with an increasing risk at higher age. In about 10% of patients with (previously unknown)

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CHD, sudden cardiac arrest following an acute myocardial infarction is the first manifestation. [5 6 8-10] In younger patients, ion channel disease, hereditary cardiomyopathies, and myocarditis are the most frequent causes of sudden cardiac arrest. [5 10-12] Unaware of this risk, unexpected deaths of young people may me misclassified as "death due to unknown cause", leading to a false-low incidence in young people. [12]

Burden of disease

Generally, people at any age may be affected by sudden cardiac arrest. Men of about 45 years of age are a group at special risk for death following sudden cardiac arrest: In these patients, sudden cardiac arrest frequently is the first manifestation of a previous unknown cardiac or cardiovascular disease. Before the incident, these patients may have been entirely healthy. [10 11] Once they have survived sudden cardiac arrest and underwent revascularization, these patients may continue their life with the same quality of life they have had before the incidence. [8 10] But to reach this goal, patients have to be admitted to hospital with ROSC, or under ongoing CPR.

Reflecting this special group, sudden cardiac arrest is a vast socio-economic burden, as these patients are in the height of their working life (and the younger the patients, the higher the burden). Beyond the social burden of losing young persons, to the pure healthcare-related costs, which are estimated $ 33 billion per year in the United States, [13] losses of earnings and productivity have to be added in order to realize the costs across the entire society. Though, the burden of premature death in years of potential life lost is greater for sudden cardiac arrest than for all individual cancers and most other leading causes of death. [14]

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What is known?

The medical problem

CPR quality is important for resuscitation success. With manual CPR, increasing fatigue of the rescuers, and frequent interruptions of compressions have been reported. Both, fatigue and interruptions decrease blood flow required for adequate myocardial and brain perfusion, which is crucial for ROSC and good neurological outcome. Chest compressions of good quality and with little interruptions have been shown to improve survival rates, compared to insufficiently performed chest compressions. [15-17]

The issues encountered with manual CPR led to the development of mechanical CPR systems which are supposed to improve CPR quality, an essential determinant for survival. [17 18] Application studies have shown that using mechanical CPR devices, higher arterial carbon dioxide levels, indicative of better tissue perfusion, can be achieved, as well as better haemodynamic conditions, thereby leading to improved coronary and cerebral blood flow. Furthermore, it has been stated that coronary and brain perfusion is the leading determinant of survival following resuscitation. Based on the premise that automated mechanical CPR provides a sustained quality of chest compressions, better outcomes might be expected from the devices. [19]

Literature search in a primary evaluation, carried out as a supplemental appraisal of a medical thesis, showed numerous studies of heterogeneous design, many of poor quality. [19 20] Three high-quality, prospective, large, multi-centre randomized controlled trials on mechanical CPR (CIRC, [21] LINC, [22] and PARAMEDIC [23]) have been published, as well as several publications from retrospective registry analyses [19 24-27]. They find inconsistent and contradictory results, leaving it unclear if manual or mechanical chest compressions during CPR may improve the outcome of patients suffering cardiac arrest. If considered more in detail, we neither have

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evidence if there is a special point of time to install a mechanical CPR device, nor if there are special situations or subgroups of patients who will particularly benefit from mechanical chest compressions. Especially, we do not know if one compression technique (vertical piston versus semi-circumferential load-distributing band) is superior over the other. [19 28]

In current CoSTR, mechanical CPR devices were introduced as potential advanced life support adjuvant. While the advantages and disadvantages are discussed, no recommendations in favour of, or against the application of these devices are given. Thus, guidelines recommendations leave it unclear if mechanical chest compressions should be aspired during CPR. [29] In detail, they neither suggest which patient may profit from automated CPR, nor in which situation, or at which point of time mechanical CPR may be beneficial.

Need for the systematic review

A recently published Cochrane Review shows several weaknesses: [28] Most important, the authors do not consider different chest compression techniques: they miss to distinguish vertical chest compressions by a piston from semi-circumferentially constricting compressions by a load-distributing band. Therefore, this meta-analysis does not distinguish different treatments, which may have different or even contrawise effects on the outcomes, which might increase between-trial heterogeneity unnecessarily.

Beyond the Cochrane review, several other meta-analyses of poor quality have been published. One review was sponsored by the manufacturer of a mechanical CPR device. [30] As this study shows a superiority of the device produced by the sponsor of the review over the competitors' device, and over mechanical CPR, the reader should be aware of COI by the authors affiliated with this company. This has been criticized, as "industry sponsored meta-analyses are more likely to recommend an intervention". [31] Moreover, this particular study examined ROSC only, but no long-term outcomes, and included both, clinical and animal models.

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Another review provides an overview of existing systematic reviews and studies and concludes that there are more studies needed for a definitive judgement. [32] Further reviews do not consider different chest compression techniques, as described above. [28 33 34] Some reviewers only search the most important database, Medline®, though under-representing publications indexed in other databases. As numerous European journals are indexed in EMBASE®, this may have led to a selection bias and an over-representation of American trials (or trials published in American Journals). As one of the main competitors has been developed in the US, and the other in Europe, this search strategy may not be representative. Beyond, we have to consider that there are fundamental differences between the American, paramedic-based, and the European, physician-staffed emergency medical service system. [30 32 35]

Adverse effects

Important adverse effects of mechanical CPR systems are injuries caused through chest compression. They comprise bone fractures (e.g. ribs, sternum, vertebrae) and organ lacerations (e.g. liver, spleen). [36] Injuries do not seem to be more frequent than through manual chest compressions, but the pattern may differ. [19 37 38] Installation of the device may dalay the further CPR process and therefore delay life-saving measures like defibrillation, and supporting measures like airway management or intravenous access. [24 39-42] And last, as with any technical device, failures may occur.

Objectives

Patients

The objective of this systematic review and meta-analysis is to detect whether adults suffering cardiac arrest may benefit from the application of automated mechanical CPR devices.

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Cardiac arrest is defined as an absence of circulation, which may be validated by evaluating signs of life, palpation of pulse over a major artery, or invasive blood pressure measurement.

Experimental intervention

Mechanical (also referred to as "automated") CPR devices deliver chest compressions independently from the user, either through vertical force by a piston, or through semi-circumferentially constricting force by a load-distributing band. Mechanical CPR devices are also referred to as "automated CPR systems", or "external cardiac compressors".

Until application of the mechanical CPR device, the patients may have received manual chest compressions, as described below.

Control intervention

Conventional manual chest compressions follow current Advanced Life Support CoSTR valid as per study period. As these guidelines summarise all the evidence for treatment recommendations and describe the best clinical practise, they are mandatory worldwide. Furthermore, manual chest compressions are generally available. Hence, manual CPR only is considered comparator.

Minimum requirements for conventional manual chest compressions include 100-120 vertical compressions per minute of approximately one third of the sternum height, performed at middle of chest (caudal third of sternum, intermammary line) in supine position. [4]

Outcomes and priorization

ROSC is one of our primary outcomes, as it is a common endpoint in resuscitation studies which can be easily obtained and does not depend on a clinical follow-up. ROSC is defined as any spontaneous pulse, "detectable by manual palpation of a major artery" at any time following resuscitation attempts measured as binary variable. [43]

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The other primary outcome is neurologic state according to the Glasgow-Pittsburgh cerebral performance category (CPC) [44], or modified Rankin score (mRS) [45]; the association between CPC and mRS is described elsewhere [46]. Neurological outcome is evaluated after 30 days, or at hospital discharge, whichever comes first. If CPC scores of 1 and 2 are only reported summarized as "intact" or "good" neurological outcome, they are treated as binary event ("good" vs. "poor"). [44] By way of derogation from our initial study registration, we consider neurologically intact survival as a primary outcome because of the essential importance of this outcome for the indivudial patients' further life.

Secondary outcomes include short- and long-term survival, resuscitation trauma, and CPR quality surrogates (table 1).

Table 1: Secondary outcomes.

Domain	Secondary outcome
Short-term survival	• alive at hospital admission • alive ≤30 days after incident • alive at hospital discharge
Long-term survival	• alive 3, 6, and 12 months after incident
Resuscitation trauma	• frequency of injuries and trauma caused by chest compressions (including skin lesions, skeletal fractures, pulmonary lesions, pneumo-, haemothorax, haemoperitoneum, organ lacerations)
CPR quality surrogates	• guideline compliance (including chest compression depth, frequency, rate, ratio) • no-flow (hands-off) time, and ratio • arterial blood pressure during chest compressions • end-tidal carbon dioxide partial pressure ($etCO_2$) with ongoing CPR

Legend to table 1: CPR = cardiopulmonary resuscitation; $etCO_2$ = end-tidal carbon dioxide.

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CPR quality surrogates may have impact on patient survival, as fatigue and interruptions decrease blood flow required for adequate myocardial and brain perfusion, which is crucial for a favourable neurological outcome. [47 48] High fractions of chest compressions have been shown to be associated with improved survival rates, compared to insufficiently performed chest compressions. [17 49-51] Arterial blood pressure is used to describe hemodynamics, as it determines cerebral and coronary perfusion. End-tidal carbon dioxide ($etCO_2$) is considered a surrogate marker for induced cardiac output and the quantity of viable cells within the organism.

Primary and secondary outcomes are reported according to the "Utstein Style", a protocol for uniform reporting of resuscitation outcomes, [43] based in part of previous reviews.

Review questions

The review of studies is designed to address the following questions:

- Is manual CPR associated with an improved primary or secondary outcome, as stated above?
- May automated CPR improve the process, quality, and safety of CPR?
- What are possible scopes of application for automated CPR devices?

METHODS AND ANALYSIS

We registered this study at the International Prospective Register of Systematic Reviews (PROSPERO), registration number CRD42017051633. Reporting of this protocol accords to the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) statement. [52] The PRISMA-P checklist is provided as an additional file. We will follow both, Cochrane Handbook [53], and MOOSE Statement [54].

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Eligibility criteria

Study characteristics

Eligible for inclusion are randomized controlled trials (RCT) and quasi-RCT comparing mechanical to manual chest compressions in cardiac arrest if at least one of the primary endpoints is available. In this context, state (published, accepted for publication, or pre-print) and mode of publication (full article or conference abstract) are irrelevant. Studies are only eligible for inclusion, if at least one of our primary endpoints is available, and if the data for the control group is published.

Population characteristics and setting

We only include studies in adult patients (≥18 years of age), as most mechanical CPR devices are not approved for children. We consider both settings, in- and out-of-hospital cardiac arrest eligible for inclusion.

Device characteristics

All mechanical CPR devices that deliver automated chest compressions are included into the systematic review to gain comprehensive insight into the topic and avoid selection bias. The devices may compress the chest independently from the user, either through vertical force (applied by a piston) or semi-circumferentially constricting force (applied by a load-distributing band). However, we conduct subgroup analysiss for each compression technique. Technical characteristics differ for each device (figure 1, table 2). [19]

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Figure 1: Automated mechanical cardiopulmonary resuscitation devices.



Legend to figure 1: From left to right: AutoPulse® (courtesy of Zoll medical, Chelmsford, USA), LUCAS® 3 (courtesy of Stryker, Kalamazoo, USA), LifeStat® 1008 (courtesy of Michigan Instruments, Grand Rapids, USA), Corpuls® CPR (courtesy of GS Elektromedizinische Geräte G. Stemple, Kaufering, Germany). Easy Pulse® (courtesy of Schiller, Baar, Switzerland), MCC® (courtesy of Resuscitation International, Scottsdale, USA). No images provided for LifeLine® ARM and X-CPR® 2.

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Table 2: Specifications of automated mechanical CPR devices.

device	AutoPulse®	LUCAS® 3	LifeStat® 1008	Corpuls® CPR	LifeLine® ARM	Easy Pulse®	MCC® Mini Chest Compressor	X-CPR® 2
manufacturer	Zoll Medical, Chelmsford, USA (prior: Revivant)	Stryker, Kalamazoo, USA (prior: Physio-Control, Jolife)	Michigan Instruments, Grand Rapids, USA	GS Elektromedizinische Geräte G. Stemple, Kaufering, Germany	Defibtech, Guilford, USA	Schiller, Baar, Switzerland	Resuscitation International, Scottsdale, USA	CU Medical Systems, Wonju, Korea (prior: Healthwell Medical)
technique	semi-circumferentially constricting load distributing band, electric driven	piston, electric driven	piston & ventilator, pneumatic driven	piston, electric driven	piston, electric driven	piston, electric driven, non-constricting band	Piston, pneumatic driven, non-constricting band	piston with semi-circumferentially constricting thoracic strap, electric driven
compression - depth	20% of thorax diameter + 0.6 / - 1.3 cm	5.3 ± 0,2 cm (4.0–5.3 cm for sternum height <18.5 cm)	0.0–8.0 cm (adjustable)	2.0–6.0 cm (adjustable)	5.3 ± 0.3 cm	4.0–5.2 cm (adjusted for chest circumference or width)	3.8 or 5.1 cm (selectable)	[no information provided]
- rate	80 ± 5 /min	102 ± 2 /min	100 ± 6 /min	80 – 120 /min (adjustable)	101 ± 1 /min	100 /min	110 ± 22 /min	[no information provided]
compression / ventilation ratio (c/v)	30:2, 15:2, or continuous compression (selectable)	30:2 or continuous compression (selectable)	30:2 or continuous compression (selectable)	30:2, 15:2, or continuous compression (selectable)	30:2 or continuous compression (selectable), updates possible	30:2 or continuous compression (selectable)	continuous compression	[no information provided]
patient specifications	chest circumference 76–130 cm chest width 25–38 cm weight ≤136 kg	sternum height 17–30 cm chest width ≤45 cm	sternum height 11–37 cm chest width ≤56 cm	sternum height 14–34 cm chest width ≤48 cm	sternum height 17–30 cm chest width ≤46 cm	chest circumference 76–135 cm chest width 22–40 cm	chest circumference 78 – 130 cm	[no information provided]
device dimensions - size	83 × 45 × 8 cm + load-distributing band & battery	58 × 33 × 26 cm (stowed) 56 × 52 × 24 cm (assembled)	57 × 46 × 19 cm (assembled)	45 × 43 × 9 cm + board (different sizes)	51 × 51 × 25 cm (stowed) 60 × 53 × 23 cm (assembled)	18 × 22 × 15 cm	20 × 16 × 15 cm + band	[no information provided]
- weight	11.6 kg including battery	8.0 kg including battery + straps	8.9 kg + gas supply	5.5 kg including battery + board (1.7–2.1 kg)	7.1 kg including battery	3.5 kg + belt with slider (0.3 kg)	2.0 kg + gas supply	[no information provided]
- average consumption	battery run-time ≈ 30 min	battery run-time ≈ 45 min external power supply	≤45 l/min compressed gas	battery run-time ≈ 90 min, rechargeable during use	battery run-time ≈ 60 min	battery run-time ≈ 45 min external power supply	60 ± 5 l/min compressed gas	[no information provided]

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features	event recorder	report generator	n/a	memory card interface	n/a	computer interface	n/a	n/a
forerunner models	**vest CPR**: inflatable vest, pneumatic driven, circumferential pressure 200-280 mmHg, compression rate 60 /min	**LUCAS®**: pneumatic driven piston, continuous compressions **LUCAS®** 2: electric driven piston	**thumper®** (under licence for Europe: Siemens **Sirepuls®**): model *1001*, *1002*, *1007CC*: no ventilator, continuous compressions *1003*, *1004*, *1005*, *1007*, *1007CCV*: C/V 5:1 or continuous compressions	n/a	n/a	n/a	n/a currently not selling in Europe & USA	**X-CPR®** (SST-CPR®), pneumatic driven piston & strap currently not selling

Legend to table 2: Specifications according to manufacturer information. Abbreviations: CPR = cardiopulmonary resuscitation; c/v = compression/ventilation ratio; n/a = not applicable; mmHg = millimetres of mercury.

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Until the first application of the automated CPR device, patients may receive manual chest compressions as described above.

Exclusion criteria

Case-series, case-control-studies, or historically controlled studies studies are excluded, as well as studies involving traumatic cardiac arrest, as treatment priorities fundamentally differ from non-traumatic cardiac arrest. We expect this population to be representative for patients suffering sudden cardiac arrest, which is predominantly of cardiac aetiology.

We do not apply any restrictions on manufacturer, model, technique, or publication language.

Information sources

We consider relevant bibliographic databases, as listed in table 3, until the search date and document the date of our last access on the search engine.

Table 3: Searched databases.

database
Journals@Ovid Full Text
Your Journals@Ovid
EBM Reviews - Cochrane Database of Systematic Reviews - ACP Journal Club - Database of Abstracts of Reviews of Effects - Cochrane Central Register of Controlled Trials - Cochrane Methodology Register - Health Technology Assessment - NHS Economic Evaluation Database
BIOSIS Previews®
Embase®
Ovid MEDLINE®

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- in-process & other non-indexed citations - daily update

Legend to table 3: EBM = evidence-based medicine; ACP = American College of Physicians; NHS = National Health Service.

Search strategy and study selection

Search strategy

Our search strategy comprises several steps: Initially, we perform a combined, structured, and keyword-based search containing search terms in the field of cardiopulmonary resuscitation, the devices' names, and techniques, as presented in table 4. Backward reference searching may identify further literature eligible for inclusion by analysing the reference sections of the articles found with our search strategy. Forward reference searching is executed to identify additional relevant studies that cite a study included. Pertinent study registries are searched via the World Health Organization (WHO) International Clinical Trials Registry Platform (ICTRP; http://apps.who.int/trialsearch) for trials, which are registered, but not published. Finally, we ask manufacturers of mechanical CPR devices for unpublished data, and assess preprint servers and data repositories.

Table 4: Keyword-based search.

#	search item
1	mechanical chest compressors
2	mechanical CPR device
3	chest compression device
4	mechanical cardiopulmonary resuscitation
5	vest CPR
6	Thumper
7	AutoPulse
8	Lund University Cardiac Assist System

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9	LUCAS-2 *or* LUCAS2 *or* LUCAS-3 *or* LUCAS3
10	LifeStat *or* Life-Stat
11	Miniaturized Chest Compressor
12	Animax
13	Corpuls CPR
14	LifeLinie ARM
15	EasyPulse
16	X-CPR
17	load distributing band
18	mechanical piston
19	resuscitation/ *or* resuscitator/ *or* advanced life support
20	1 *or* 2 *or* 3 *or* 4 *or* 5 *or* 6 *or* 7 *or* 8 *or* 9 *or* 10 *or* 11 *or* 12 *or* 13 *or* 14 *or* 15 *or* 16 *or* 17 *or* 18
21	19 *and* 20
22	*remove duplicates from* 21

Legend to table 4: Multiple entries are feasible. CPR = cardiopulmonary resuscitation.

For literature search, we use the proprietary search engine Ovid®, Ovid Technologies Inc., Wolters Kluwer Health, New York/USA.

Study selection

Initially, we screen both, title and abstract of the publications identified through our search, for eligibility. If the study meets any inclusion, but no exclusion criteria, we assess the full text in order to extract study characteristics, results, and risks of bias, as stated below.

Through our search strategy (table 4), duplicates are removed automatically. Nevertheless, additional duplicates may occur if results of one study are published in several manuscripts with different titles, in different languages, or different modes of publication. We consider publications duplicates if the same authors, or authors from the same centres, publish studies covering the same observation period, with comparable case numbers, similar design, and

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related study objectives. In case of redundant publication, we prefer full texts to abstracts, and merge the data published.

Study records

Data management

Characteristics and results from the studies included are extracted and summarized in a table (Numbers® 6.2, Apple Inc., Cupertino/USA). If eligible, frequencies of primary and secondary outcomes are charted in forest plots (Review Manager® 5.3, The Nordic Cochrane Centre, Copenhagen/Denmark).

Selection process and Collection of data

Study selection, quality assessment, and data extraction is performed independently by two reviewers of which at least one with clinical experience in resuscitation and one with methodological experience. Possible disagreements and discrepancies are resolved by consensus, or by consulting a third reviewer.

We contact the corresponding authors of included studies to obtain key missing data as needed, or to clarify any issues and questions on a particular trial. If authors do not response within two weeks, we contact the corresponding author again, as well as first and senior authors. If this does not lead to an elaboration of the issue, we exclude any outcomes from further analysis whose context cannot be clarified. We only include results into our meta-analysis if their origin is transparent and accurate.

Data items

Data is extracted using a pre-defined data collection form according to the Cochrane Collaboration recommendations (table 5). [55]

Table 5: Data extraction.

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Domain	Data items
Unambiguous study identification	• authors (up to three), year, title, journal or other source, volume, page(s) • identification as conference abstract, or research letter, if applicable • identification number within a study registry, if applicable
Study design	• study type • chronological dimension (prospective, retrospective) • number of study centres (single-centre for 1; multi-centre for ≥5 centres) • characteristics of patients included • type of intervention(s) • case number in total, distributed among intervention and control groups, and subgroups if applicable • mechanical CPR system(s) examined • CoSTR version the study is based on
Results	• qualitative summary • quantitative summary
Risks of bias	• risks of bias according to the Cochrane tool • funding sources • conflicts of interest

Legend to table 5: Data collected from included studies.

Assumptions and simplifications

Some authors indicate piston-driven mechanical chest compressions as "manual CPR", particularly when comparing them to compressions by a load-distributing band. Regardless of the denomination chosen by the authors, we only consider chest compressions delivered by hand as "manual CPR".

Cardiac arrest is defined as the absence of cardiovascular circulation. This condition may be validated by evaluating signs of life, palpation of pulse over a major artery, or invasive blood pressure measurement.

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If the underlying CoSTR is not stated within the text, we collate the trial to the CoSTR valid as per study period.

In some publications, "hands-off time" (in seconds) or "hands-off ratio" (in percentage of "hands-on" or compression time) may be indicated in lieu of "no-flow time" (in seconds).

Risk of bias in individual studies

For each included RCT, we follow Cochrane Risk of Bias tool, which is the standard tool for assessing risks of bias in RCTs. [53] Each of the following potential sources of bias is evaluated where applicable:

- random sequence generation
- allocation concealment
- blinding of participants and personnel
- blinding of outcome assessment
- incomplete outcome data
- selective outcome reporting
- other bias (including baseline imbalance, early termination of the trial, funding bias, etc.)

Each is graded as "high", "low", or "unclear", and a justification for the judgment is presented in the risk of bias table.

We assess the risk of bias for included quasi-RCT with Cochrane ROBINS-I tool. [56] The following categories are evaluated at study level:

- confounding
- selection of participants into the study
- classification of interventions

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- deviations from intended interventions
- missing data
- measurement of outcomes
- selection of the reported result

Each domain is categorized "critical", "serious", "moderate", "low" risk of bias, or "no information". Again, we present justification of the judgement in the risk of bias table. The overall risk of bias for one outcome of quasi-RCT is determined by the highest risk of bias rating of the particular item.

Data synthesis and analysis

Statistical analytics

For dichotomous outcomes (primary and secondary outcomes), odds ratio (OR) with its 95% confidence interval (CI) is used as effect measure per trial. In binary outcomes where rare or 0 events are expected in some trial-arms (i.e. some of the secondary outcomes as trauma), the Peto OR is used to avoid the necessity of a continuity correction. As the Peto OR is only suitable for rare events [53], conventional OR are used for all other binary outcomes. If possible, the OR and its standard error is extracted directly, preferably from an adjusted model. If it is not reported, but the events and numbers of patients per intervention group are available, then the unadjusted OR along with its standard error is obtained. For continuous outcomes, the standardized mean difference (SMD) with its 95% CI is used as effect measure per trial. For survival outcomes, (adjusted) HR estimated in Cox regression models are used if reported. Results extracted from Log-rank tests, Kaplan-Meier curves or reported Kaplan-Meier estimates are used alternatively. If neither Cox-regression nor Kaplan-Meier estimators are reported, survival outcomes are extracted as dichotomous outcome. ORs are combined using the Mantel-Haenszel method, and SMDs are pooled by the inverse-variance method. Effect

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measures are combined separated by study design (RCT and quasi-RCT). [53] Random-effects models are used to account for the expected between-trial heterogeneity in estimating the overall effect and the Hartung-Knapp method is applied for the corresponding CI, if at least 5 trials contribute information to an estimate. [57] Results of fixed-effect models are provided and discussed in contrast to the results of the random-effects models. We investigate all results for statistical heterogeneity by I^2-statistics and by interpreting it in relation to the estimated between-trial variance τ^2. If there is a considerable heterogeneity ($I^2 \geq 75\%$), no meta-analysis is performed.

R version 3.5.0 or higher (R Foundation for Statistical Computing, Vienna, Austria) and the R meta package version 4.9-5 or higher (developed by Guido Schwarzer) are used for all statistical analyses. For generating diagrams, we utilize Review Manager®, Version 5.2.7 (The Nordic Cochrane Centre, Copenhagen, Denmark).

Meta-bias

We perform statistical investigation of potential publication bias based on a test of funnel plot asymmetry, if there are at least 10 studies pooled for one outcome. Sensitivity analyses are conducted with regard to the studies' quality. Further sensitivity analyses are performed during the course of the review, if irregularities of individual studies are found. We set sup "Summary of findings" as recommended by Cochrane Collaboration. [53 55]

In order to detect reporting bias, we compare the data reported in publications with a priori study protocols, or registrations.

Confidence in cumulative evidence

Quality of evidence is analysed with GRADEpro® GDT (Guideline Development Tool, Grade Working Group, McMaster University and Evidence Prime Inc., Hamilton, Canada). GRADE

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is an accepted tool for quality assessment and is used to grade the body of evidence at outcome level. [58]

Interventions

Titles and/or abstracts of studies retrieved using the search strategy and those from additional sources after tracing reference sections are screened to identify studies that potentially meet the inclusion criteria. The full text of those potentially eligible studies is retrieved and assessed for eligibility. We use a standardised questionnaire to extract data from the included studies for assessment of study quality and evidence synthesis.

Qualitative synthesis

We provide a narrative synthesis of the findings from the studies included, structured around the mechanical chest compression technique, and summarize study characteristics, including risks of bias and outcomes, in tables.

Quantitative synthesis

Whenever possible, we provide summaries of the intervention effects by using the effect measures along with their 95% CI and p-values for each trial and for the combined effect. The results are graphically illustrated by forest plots.

Subgroups

We plan subgroup analyses for the two different techniques of mechanical chest compression (semi-circumferentially constricting load-distributing band vs. vertically compressing mechanical piston); for the setting (urban vs. rural, and Europe vs. United States of America vs. other regions); as well as for in- and out-of-hospital cardiac arrest. If applicable, we perform subgroup analysis of aetiology, and concomitant treatment.

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Heterogeneity

All results are investigated for statistical heterogeneity by I^2-statistics and by the between-trial variance τ^2. [59] If there is a considerable heterogeneity ($I^2 \geq 75\%$) for an outcome, no meta-analysis will be performed.

Sensitivity analysis is performed to identify scenarios (like non-compliance, protocol violations, missing data, outcome definitions, clustering or correlation, competing risks, baseline imblalance, distibrutional assumptions, or outlies) that influence the findings of the review. [60] Results of sensitivity analysis are presented with the study assessment sheets.

Amendments

Any protocol amendment will be stated with reason within the subsequent publications' "methods" section.

Patient and public involvement

Unfortunately, there are no organisations of patients or their relatives after cardiac arrest in Germany. As with mortality rates of 80-90% only a minority survives the incident, [5-7] and even less with neurologically intact survival. Therefore, and unlike to patients suffering from chronical diseases with creeping progressions, it is characteristic of critical care medicine that patients cannot be involved in projects and organizations during an emergency. This applies particularly to resuscitation studies with high mortality rates and survivors requiring immediate intensive care treatment. Therefore, our focus of patient involvement in these cases is on prevention and education.

All cooperating institutions are involved in the organization or particular projects of the "Week of Resuscitation" ("Woche der Wiederbelebung"), an initiative hosted annually in the middle of September, highlighting the importance of recognising sudden cardiac arrest and

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delivering proper CPR. [7] By these events, we try to get in touch with surviving patients or dependants of cardiac arrest and to assess their expectations towards CPR and outcome.

By defining survival with neurologically intact outcome as primary outcome, we put our focus on the outcome which is most important to the patient. This topic is of specific interest of our patients, because the quality of CPR in general and the quality of chest compressions in particualr will determine their abilities or disabilities, and their quality of life after the incident (i. e. the "neurological outcome"). Unfortunately, this constitutes an endpoint which is not easy to assess and therefore is uncommon in resuscitation studies.

ETHICS AND DISSEMINATION

No further ethical approval is needed because data from previous studies is retrieved and analysed, whose authors declared clearance by the responsible ethic commissions.

As given by their nature, resuscitation studies cannot meet one crucial issue of the Declaration of HELSINKI Ethical Principles for Medical Research Involving Human Subjects [61] and are regularly conducted under an emergency exception for informed consent. [62-65] Particularly with regard to this predicament, implementation of ethical considerations in resuscitation studies has to be assessed critically.

The publication contains data deriving from a preliminary project as a part of a doctoral thesis at Ulm University. [19 20]

Upon completion, we intend to prepare a manuscript for publication in an international peer-reviewed journal and present the results at scientific conferences.

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DISCUSSION

A primary evaluation of this work identified 107 trials on the comparison of mechanical and manual CPR on 179,505 patients, 444 pigs, 62 dogs, and 723 manikin scenarios. This evaluation did not meet established quality criteria of a systematic review, but we were able to become aware of the issues relied with mechanical CPR, and current evidence. [19 20]

Mechanical CPR devices are considered potential advanced life support adjuvants. While their advantages and disadvantages are discussed, no recommendations in favour of or against the application of these devices are provided. Though, guidelines leave it unclear if mechanical chest compressions should be aspired during CPR. In detail, they neither suggest which patient may profit from automated CPR, nor in which situation, or at which point of time mechanical CPR may be beneficial. [29 66]

Strenghts and Limitations

When comparing mechanical to manual CPR, several confounders may influence the patients' outcome, which in their entirety are complex to detect. Any prospective study may be subject to the Hawthorne Effect, when participants show a different behaviour due to the knowledge that they participate in a study. The first automated CPR systems have been developed just shortly after the description of manual chest compression in the 1960s. [67] Therefore, studies over half of a century have to be assessed, and revisions of the CoSTR have to be considered in the meantime. On the other hand, it is yet to determine to what extent study designs comply with particular CoSTR recommendations.

Publication bias is a common problem in evidence-based medicine, as only studies may be assessed if the authors are aware of their existence. In order to reduce publication bias, we ask shareholders for unpublished data, but risk of publication bias will depend from their cooperativeness.

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Conference abstracts typically are subject to word limits, impairing the possibility to respond to all aspects of study design and conduction in a short text. To prevent bias caused by lacking information from abstracts, we ask the authors for additional information. Again, prevention of bias requires their cooperativeness.

Surrogate outcomes like ROSC or survival rate until hospital admission are easier to obtain than "hard" clinical outcomes which require considerable resources for study design, funding, recruitment, conduction, and follow-up. But patients are neither interested in survival rates nor in rates of ROSC achieved – the relevant outcome from our patients' point of view is at which neurological state they survive. Hence, neurologically intact survival should be considered of most particular interest.

LIST OF ABBREVIATIONS

τ^2	between-trial variance
ACP	American College of Physicians
C/V	compression/ventilation ratio
CI	confidence interval
CoSTR	Consensus on Cardiopulmonary Resuscitation and Emergency Cardiovascular Care Science with Treatment Recommendations
CPC	GLASGOW-PITTSBURGH cerebral performance category
CPR	cardiopulmonary resuscitation
EBM	evidence-based medicine
$etCO_2$	end-tidal carbon dioxide
I^2	inconsistency
MAP	mean arterial pressure

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mmHg	millimetres of mercury
mRS	modified RANKIN Score
n/a	not applicable
NHS	National Health Service
OR	odds ratio
ROSC	return of spontaneous circulation
SD	standard deviation

DECLARATIONS

Author Contributions

Study conception and design: MO, JBZ, CMM, KJ, SES

Acquisition of data: not applicable for study protocols

Data analysis and interpretation: not applicable for study protocols

Drafting of the manuscript: MO, SES

Critical revision for important intellectual content: JBZ, EP, MAW, SW, ADL, CMM, BLN, JTG, SS

All authors read and approved the final manuscript. They agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. MO will have full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

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Competing interests

The authors declare that they have no competing interests.

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681 Stroke Foundation of Canada, InterAmerican Heart Foundation, Resuscitation Council of
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		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	title page
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	n/a
Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	8
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	title page
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	23
Amendments			

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	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	n/a; 19
Support			
Sources	#5a	Indicate sources of financial or other support for the review	24
Sponsor	#5b	Provide name for the review funder and / or sponsor	24
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	n/a
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is already known	1-2
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	5-8
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	9-10
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	10-11
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	11-12
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	12-13
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	13
Study records - data collection process	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for	13

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		obtaining and confirming data from investigators	
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	13-14
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	6-8
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	15-16
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	18
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	16-18
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	18-19
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	n/a
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	17
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	17-18

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BMJ Open

Automated mechanical cardiopulmonary resuscitation devices versus manual chest compressions in the treatment of cardiac arrest: Protocol of a systematic review and meta-analysis comparing machine to human

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Automated mechanical cardiopulmonary resuscitation devices versus manual chest compressions in the treatment of cardiac arrest:

Protocol of a systematic review and meta-analysis comparing machine to human

TITLE PAGE

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ABSTRACT

Introduction:

Cardiac arrest is a leading cause of death in industrialized countries. Cardiopulmonary Resuscitation (CPR) guidelines follow the principles of closed chest compression as described in 1960. Mechanical CPR devices are designed to improve chest compression quality, thus considered to improve resuscitation outcomes.

This protocol outlines a systematic review and meta-analysis methodology to assess trials investigating the therapeutic effect of automated mechanical CPR devices on the rate of Return of Spontaneous Circulation, neurological state, and secondary endpoints (including short- and long-term survival, injuries, and surrogate parameters for CPR quality), compared to manual chest compressions in adults with cardiac arrest.

Methods and Analysis:

A sensitive search strategy is employed in established bibliographic databases from inception until the date of search, followed by forward and backward reference searching. Into qualitative analysis, we include randomized and quasi-randomized trials comparing mechanical to manual CPR. Into qualitative analysis, studies reporting any primary endpoint are included. Case series, case control-studies, and longitudinal cohort studies are excluded from meta-analysis.

Two reviewers independently assess publications using a pre-defined data collection form according to Cochrane recommendations. Standardized tools are used for data extraction, risks of bias and quality of evidence are assessed using the Cochrane Risk-of-Bias (RoB), ROBINS-I, and GRADE tools. If enough studies are identified for meta-analysis, the measures of association are calculated by dint of bivariate random-effects models. Statistical

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heterogeneity is evaluated with I^2 statistics and explored through sensitivity analysis. Reporting follows the PRISMA statement.

Ethics and dissemination:

No ethical approval will be needed because data from previous studies in will be retrieved and analysed. Most resuscitation studies are conducted under an emergency exception for informed consent. The publication contains data deriving from a dissertation project at Ulm University. We will disseminate the results through publication in a peer-reviewed journal and scientific conferences.

Registration:

PROSPERO CRD42017051633

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ARTICLE SUMMARY

Strengths and limitations of this study

- This systematic review and meta-analysis addresses gaps in the current evidence-base on chest compression techniques as part of cardiopulmonary resuscitation by providing a comprehensive assessment of randomized and quasi-randomized clinical trials, and comprehensive subgroup analysis.
- Exensive and differentiated multi-step search strategy including 12 databases without language restrictions.
- Methods follow the Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) Protocols guidelines, Meta-Analysis Of Observational Studies in Epidemiology (MOOSE) Statement, and Cochrane Handbook.
- This review is limited to randomized and quasi-randomized clinical trials of heterogeneous design and quality.
- Potential clarification which subpopulations in which situations, under which circumstances, and at which time may benefit profit from which chest compression technique may improve threatment decisions.

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INTRODUCTION

For thousands of years, mankind has struggled to "re-animate" people showing no signs of life. A milestone of Cardiopulmonary Resuscitation (CPR), manual closed-chest cardiac massage, was introduced in 1960. [1] More recent developments are machines that perform chest compressions mechanically and automatically through inflatable vests, pistons, or load distributing bands. Can automated mechanical chest compression devices surpass human performance and again revolutionise CPR?

Rationale

Many young and able-bodied patients who suffer sudden cardiac arrest have the potential for recovery with good quality of life, but this recovery hinges on high quality CPR at the time of the event. [2-4]

Burden of disease

Sudden cardiac arrest following coronary heart disease is a major cause of death in industrialized nations with an age-relaed prevalence of 80-150 per 100,000 person years and mortality rates as high as 80-90%. In about 10% of patients, sudden cardiac arrest following an acute myocardial infarction is the first manifestation of a (previously unknown) corornary heart disease. In younger patients, ion channel disease, hereditary cardiomyopathies, and myocarditis are frequent causes of sudden cardiac arrest. [2-9]

Reflecting this special group, sudden cardiac arrest is a vast socio-economic burden, as these patients are in the height of their working life (and the younger the patients, the higher the burden). Beyond the social burden of losing young persons, to the pure healthcare-related costs, which are estimated $ 33 billion per year in the United States, [10] losses of earnings and productivity have to be added in order to realize the costs across the entire society. Though,

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the burden of premature death in years of potential life lost is greater for sudden cardiac arrest than for all individual cancers and most other leading causes of death. [11]

What is known?

The medical problem

CPR quality is important for resuscitation success. With manual CPR, increasing fatigue of the rescuers, and frequent interruptions of compressions have been reported. Both, fatigue and interruptions decrease blood flow required for adequate myocardial and brain perfusion, which is crucial for ROSC and good neurological outcome. Chest compressions of good quality and with little interruptions have been shown to improve survival rates, compared to insufficiently performed chest compressions. [12-14]

The issues encountered with manual CPR led to the development of mechanical CPR systems which are supposed to improve CPR quality, an essential determinant for survival. [14 15] Application studies have shown that using mechanical CPR devices, higher arterial carbon dioxide levels, indicative of better tissue perfusion, can be achieved, [16-18] as well as better haemodynamic conditions, thereby leading to improved coronary and cerebral blood flow. [16 19-21] Furthermore, it has been stated that coronary and brain perfusion is the leading determinant of survival following resuscitation. Based on the premise that automated mechanical CPR provides a sustained quality of chest compressions, better outcomes might be expected from the devices. [22]

Literature search in a primary evaluation, carried out as a supplemental appraisal of a medical thesis, showed numerous studies of heterogeneous design, many of poor quality. [22 23] Three high-quality, prospective, large, multi-centre randomized controlled trials on mechanical CPR (CIRC, [24] LINC, [25] and PARAMEDIC [26]) have been published, as well as several publications from retrospective registry analyses [22 27-30]. They find inconsistent and contradictory results,

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leaving it unclear if manual or mechanical chest compressions during CPR may improve the outcome of patients suffering cardiac arrest. If considered more in detail, we neither have evidence if there is a special point of time to install a mechanical CPR device, nor if there are special situations or subgroups of patients who will particularly benefit from mechanical chest compressions. Especially, we do not know if one compression technique (vertical piston versus semi-circumferential load-distributing band) is superior over the other. [22 31]

Need for the systematic review

A recently published Cochrane Review shows several weaknesses: [31] Most important, the intended subgroup analysis to distinguish vertical chest compressions by a piston from semi-circumferentially constricting compressions by a load-distributing band, and in- from out-of-hospital cardiac arrest was abandoned due to the variability of included trials. Therefore, this meta-analysis does not consider different treatments, which may have different or even contrawise effects on the outcomes, which might increase between-trial heterogeneity unnecessarily.

Beyond the Cochrane review, several other meta-analyses of poor quality have been published. One review was sponsored by the manufacturer of a mechanical CPR device. [32] As this study shows a superiority of the device produced by the sponsor of the review over the competitors' device, and over mechanical CPR, the reader should be aware of conflicts of interest by the authors affiliated with this company. This has been criticized, as "industry sponsored meta-analyses are more likely to recommend an intervention". [33] Moreover, this particular study examined ROSC only, but no long-term outcomes, and included both, clinical and animal models. Potential conflicts of interest may also be considered if authors conducting the systematic review co-authored a trial which is included into the review, as they have to critically assess their own publications. [24 26 34 35]

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Another review provides an overview of existing systematic reviews and studies and concludes that there are more studies needed for a definitive judgement. [36] Systematic reviews are at increased risk of selection and attrition bias and confounding if observational studies are included, [37] which applies to some reviews. [22 32 35 38] Further reviews do not distinguish different chest compression techniques in their meta-analysis, as described above, [31 35 38 39] or only include one particular device. [40-42] Others focus only on in- [43] or out-of-hospital [32 34 35 39 41 42] cardiac arrest, respectively.

Some reviewers only search the most important database, Medline®, though under-representing publications indexed in other databases. As numerous European journals are indexed in EMBASE®, this may have led to a selection bias and an over-representation of American trials (or trials published in American Journals). As one of the main competitors has been developed in the US, and the other in Europe, this search strategy may not be representative. Beyond, we have to consider that there are fundamental differences between the American, paramedic-based, and physician-staffed emergency medical service systems as largely provided in European countries. [32 36 44]

Adverse effects

Important adverse effects of mechanical CPR systems are injuries caused through chest compression. They comprise bone fractures (e.g. ribs, sternum, vertebrae) and organ lacerations (e.g. liver, spleen). [45] Injuries do not seem to be more frequent than through manual chest compressions, but the pattern may differ. [22 46 47] Installation of the device may delay the further CPR process and therefore delay life-saving measures like defibrillation, and supporting measures like airway management or intravenous access. [27 48-51] And last, as with any technical device, failures may occur.

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Objectives

Patients

The objective of this systematic review and meta-analysis is to detect whether adults suffering cardiac arrest, or specific subpopulations may benefit from the application of automated mechanical CPR devices.

Experimental intervention

Mechanical CPR devices deliver chest compressions independently from the user, either through vertical force by a piston, or through semi-circumferentially constricting force by a load-distributing band. So far, it is unknown if one compression technique might be superior. Mechanical CPR devices are also referred to as "automated CPR systems", or "external cardiac compressors".

Until application of the mechanical CPR device, the patients may have received manual chest compressions, as described below. To date we steill do not know if the timing of the devices' application is of importance.

Control intervention

Conventional manual chest compressions follow valid recommendations. As these guidelines summarise all the evidence for treatment recommendations and describe the best clinical practise, they are mandatory worldwide. Furthermore, manual chest compressions are generally available. Hence, manual CPR only is considered comparator.

Minimum requirements for conventional manual chest compressions include 100-120 vertical compressions per minute of approximately one third of the sternum height, performed at middle of chest (caudal third of sternum, intermammary line) in supine position. [52]

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Outcomes and priorization

ROSC is one of our primary outcomes, as it is a common endpoint in resuscitation studies which can be easily obtained and does not depend on a clinical follow-up. ROSC is defined as any spontaneous pulse, "detectable by manual palpation of a major artery" at any time following resuscitation attempts measured as binary variable. [53]

The other primary outcome is neurologic state according to the Glasgow-Pittsburgh cerebral performance category (CPC) [54], or modified Rankin score (mRS) [55]; the association between CPC and mRS is described elsewhere [56]. Neurological outcome is evaluated after 30 days, or at hospital discharge, whichever comes first. If CPC scores of 1 and 2 are only reported summarized as "intact" or "good" neurological outcome, they are treated as binary event ("good" vs. "poor"). [54] By way of derogation from our initial study registration, we consider neurologically intact survival as a primary outcome because of the essential importance of this outcome for the indivudial patients' further life.

Secondary outcomes include short- and long-term survival, resuscitation trauma, and CPR quality surrogates (table 1).

Table 1: Secondary outcomes.

Domain	Secondary outcome
Short-term survival	• alive at hospital admission • alive ≤30 days after incident • alive at hospital discharge
Long-term survival	• alive 3, 6, and 12 months after incident
Resuscitation trauma	• frequency of injuries and trauma caused by chest compressions (including skin lesions, skeletal fractures, pulmonary lesions, pneumo-, haemothorax, haemoperitoneum, organ lacerations)
CPR quality surrogates	• guideline compliance (including chest compression depth,

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	frequency, rate, ratio) • no-flow (hands-off) time, and ratio • arterial blood pressure during chest compressions • end-tidal carbon dioxide partial pressure ($etCO_2$) with ongoing CPR

Legend to table 1: CPR = cardiopulmonary resuscitation; $etCO_2$ = end-tidal carbon dioxide.

CPR quality surrogates may have impact on patient survival, as fatigue and interruptions decrease blood flow required for adequate myocardial and brain perfusion, which is crucial for a favourable neurological outcome. [57 58] High fractions of chest compressions have been shown to be associated with improved survival rates, compared to insufficiently performed chest compressions. [14 59-61] Arterial blood pressure is used to describe hemodynamics, as it determines cerebral and coronary perfusion. End-tidal carbon dioxide ($etCO_2$) is considered a surrogate marker for induced cardiac output and the quantity of viable cells within the organism.

Primary and secondary outcomes are reported according to the "Utstein Style", a protocol for uniform reporting of resuscitation outcomes, [53] based in part of previous reviews.

Review questions

The review of studies is designed to address the following questions:

- Is a specific CPR technique, manual or mechanical (by band or piston) associated with an improved primary or secondary outcome, as stated above?
- May automated mechanical CPR devices improve the process, quality, and safety of CPR?
- What are possible scopes of application for automated mechanical CPR devices? This question seeks to ascertain which patient subgroups in which situations, under which circumstances, and at which time may benefit from automated mechanical CPR.

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METHODS AND ANALYSIS

We registered this study at the International Prospective Register of Systematic Reviews (PROSPERO), registration number CRD42017051633. Reporting of this protocol accords to the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) statement. [62] The PRISMA-P checklist is provided as an additional file. We will follow both, Cochrane Handbook [63], and MOOSE Statement [64].

Eligibility criteria

Study characteristics

Eligible for inclusion are randomized controlled trials (RCT) and quasi-RCT comparing mechanical to manual chest compressions in cardiac arrest if at least one of the primary endpoints is available. In this context, state (published, accepted for publication, or pre-print) and mode of publication (full article or conference abstract) are irrelevant. Studies are only eligible for inclusion, if at least one of our primary endpoints is available, and if the data for the control group is published.

Population characteristics and setting

We only include studies in adult patients (≥18 years of age), as most mechanical CPR devices are not approved for children. We consider events "in-hospital" if the cardiac arrest occurs in a hospitalized patient, and the institutions' Medical Emergency Team is activated. Events treated by official Emergency Medical Services outside hospital builings are considered "out-of-hospital". Both settings, in- and out-of-hospital cardiac arrest, are eligible for inclusion.

Device characteristics

All mechanical CPR devices that deliver automated chest compressions are included into the systematic review to gain comprehensive insight into the topic and avoid selection bias. We identified available devices by a preliminary study [22 23] and subsequent internet search. The

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devices may compress the chest independently from the user, either through vertical force (applied by a piston) or semi-circumferentially constricting force (applied by a load-distributing band). However, we conduct subgroup analysis for each compression technique. Technical characteristics differ for each device (table 2). [22]

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Table 2: Specifications of automated mechanical CPR devices.

device	AutoPulse®	LUCAS® 3	LifeStat® 1008	Corpuls® CPR	LifeLine® ARM	Easy Pulse®	MCC® Mini Chest Compressor	X-CPR® 2
manufacturer	Zoll Medical, Chelmsford, USA (prior: Revivant)	Stryker, Kalamazoo, USA (prior: Physio-Control, Jolife)	Michigan Instruments, Grand Rapids, USA	GS Elektromedizinische Geräte G. Stemple, Kaufering, Germany	Defibtech, Guilford, USA	Schiller, Baar, Switzerland	Resuscitation International, Scottsdale, USA	CU Medical Systems, Wonju, Korea (prior: Healthwell Medical)
technique	semi-circumferentially constricting load distributing band, electric driven	piston, electric driven	piston & ventilator, pneumatic driven	piston, electric driven	piston, electric driven	piston, electric driven, non-constricting band	Piston, pneumatic driven, non-constricting band	piston with semi-circumferentially constricting thoracic strap, electric driven
compression - depth	20% of thorax diameter + 0.6 / - 1.3 cm	5.3 ± 0,2 cm (4.0–5.3 cm for sternum height <18.5 cm)	0.0–8.0 cm (adjustable)	2.0–6.0 cm (adjustable)	5.3 ± 0.3 cm	4.0–5.2 cm (adjusted for chest circumference or width)	3.8 or 5.1 cm (selectable)	[no information provided]
- rate	80 ± 5 /min	102 ± 2 /min	100 ± 6 /min	80 – 120 /min (adjustable)	101 ± 1 /min	100 /min	110 ± 22 /min	[no information provided]
compression / ventilation ratio (c/v)	30:2, 15:2, or continuous compression (selectable)	30:2 or continuous compression (selectable)	30:2 or continuous compression (selectable)	30:2, 15:2, or continuous compression (selectable)	30:2 or continuous compression (selectable), updates possible	30:2 or continuous compression (selectable)	continuous compression	[no information provided]
patient specifications	chest circumference 76–130 cm chest width 25–38 cm weight ≤136 kg	sternum height 17–30 cm chest width ≤45 cm	sternum height 11–37 cm chest width ≤56 cm	sternum height 14–34 cm chest width ≤48 cm	sternum height 17–30 cm chest width ≤46 cm	chest circumference 76–135 cm chest width 22–40 cm	chest circumference 78 – 130 cm	[no information provided]
device dimensions - size	83 × 45 × 8 cm + load-distributing band & battery	58 × 33 × 26 cm (stowed) 56 × 52 × 24 cm (assembled)	57 × 46 × 19 cm (assembled)	45 × 43 × 9 cm + board (different sizes)	51 × 51 × 25 cm (stowed) 60 × 53 × 23 cm (assembled)	18 × 22 × 15 cm	20 × 16 × 15 cm + band	[no information provided]
- weight	11.6 kg including battery	8.0 kg including battery + straps	8.9 kg + gas supply	5.5 kg including battery + board (1.7–2.1 kg)	7.1 kg including battery	3.5 kg + belt with slider (0.3 kg)	2.0 kg + gas supply	[no information provided]
- average consumption	battery run-time ≈ 30 min	battery run-time ≈ 45 min external power supply	≤45 l/min compressed gas	battery run-time ≈ 90 min, rechargeable during use	battery run-time ≈ 60 min	battery run-time ≈ 45 min external power supply	60 ± 5 l/min compressed gas	[no information provided]

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features	event recorder	report generator	n/a	memory card interface	n/a	computer interface	n/a	n/a
forerunner models	**vest CPR**: inflatable vest, pneumatic driven, circumferential pressure 200-280 mmHg, compression rate 60 /min	**LUCAS®**: pneumatic driven piston, continuous compressions **LUCAS®** 2: electric driven piston	**thumper®** (under licence for Europe: Siemens **Sirepuls®**): model *1001*, *1002*, *1007CC*: no ventilator, continuous compressions *1003*, *1004*, *1005*, *1007*, *1007CCV*: C/V 5:1 or continuous compressions	n/a	n/a	n/a	n/a currently not selling in Europe & USA	**X-CPR®** (SST-CPR®), pneumatic driven piston & strap currently not selling

Legend to table 2: Specifications according to manufacturer information. Abbreviations: CPR = cardiopulmonary resuscitation; c/v = compression/ventilation ratio; n/a = not applicable; mmHg = millimetres of mercury.

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Until the first application of the automated CPR device, patients may receive manual chest compressions as described above.

Exclusion criteria

Case-series, case-control-studies, or historically controlled studies studies are excluded, as well as studies involving traumatic cardiac arrest, as treatment priorities fundamentally differ from non-traumatic cardiac arrest. We expect this population to be representative for patients suffering sudden cardiac arrest, which is predominantly of cardiac aetiology.

We do not apply any restrictions on manufacturer, model, technique, or publication language.

Information sources

We consider relevant bibliographic databases, as listed in table 3, until from inception until the date of search and document the date of our last access on the search engine.

Table 3: Searched databases.

database
Journals@Ovid Full Text
Your Journals@Ovid
EBM Reviews - Cochrane Database of Systematic Reviews - ACP Journal Club - Database of Abstracts of Reviews of Effects - Cochrane Central Register of Controlled Trials - Cochrane Methodology Register - Health Technology Assessment - NHS Economic Evaluation Database
BIOSIS Previews®
Embase®
Ovid MEDLINE®

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- in-process & other non-indexed citations
- daily update

Legend to table 3: EBM = evidence-based medicine; ACP = American College of Physicians; NHS = National Health Service.

Search strategy and study selection

Search strategy

Our search strategy comprises several steps: Initially, we perform a combined, structured, and keyword-based search containing search terms in the field of cardiopulmonary resuscitation, the devices' names, and techniques, as presented in table 4. Backward reference searching may identify further literature eligible for inclusion by analysing the reference sections of the articles found with our search strategy. Forward reference searching is executed to identify additional relevant studies that cite a study included. Pertinent study registries are searched via the World Health Organization (WHO) International Clinical Trials Registry Platform (ICTRP; http://apps.who.int/trialsearch) for trials, which are registered, but not published. Finally, we ask manufacturers of mechanical CPR devices for unpublished data, and assess preprint servers and data repositories.

Table 4: Keyword-based search.

#	search item
1	mechanical chest compressors
2	mechanical CPR device
3	chest compression device
4	mechanical cardiopulmonary resuscitation
5	vest CPR
6	Thumper
7	AutoPulse
8	Lund University Cardiac Assist System

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9	LUCAS-2 *or* LUCAS2 *or* LUCAS-3 *or* LUCAS3
10	LifeStat *or* Life-Stat
11	Miniaturized Chest Compressor
12	Animax
13	Corpuls CPR
14	LifeLinie ARM
15	EasyPulse
16	X-CPR
17	load distributing band
18	mechanical piston
19	resuscitation/ *or* resuscitator/ *or* advanced life support
20	1 *or* 2 *or* 3 *or* 4 *or* 5 *or* 6 *or* 7 *or* 8 *or* 9 *or* 10 *or* 11 *or* 12 *or* 13 *or* 14 *or* 15 *or* 16 *or* 17 *or* 18
21	19 *and* 20
22	*remove duplicates from* 21

Legend to table 4: Multiple entries are feasible. CPR = cardiopulmonary resuscitation; a slash indicates Medical Subject Headings (MeSH) terms.

For literature search, we use the proprietary search engine Ovid®, Ovid Technologies Inc., Wolters Kluwer Health, New York/USA.

Study selection

Initially, we screen both, title and abstract of the publications identified through our search, for eligibility. If the study meets any inclusion, but no exclusion criteria, we assess the full text in order to extract study characteristics, results, and risks of bias, as stated below.

Through our search strategy (table 4), duplicates are removed automatically. Nevertheless, additional duplicates may occur if results of one study are published in several manuscripts with different titles, in different languages, or different modes of publication. We consider publications duplicates if the same authors, or authors from the same centres, publish studies covering the same observation period, with comparable case numbers, similar design, and

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related study objectives. In case of redundant publication, we prefer full texts to abstracts, and merge the data published.

Study records

Data management

Characteristics and results from the studies included are extracted and summarized in a table (Numbers® 6.2, Apple Inc., Cupertino/USA). If eligible, frequencies of primary and secondary outcomes are charted in forest plots (Review Manager® 5.3, The Nordic Cochrane Centre, Copenhagen/Denmark).

Selection process and Collection of data

Study selection, quality assessment, and data extraction is performed independently by two reviewers of which at least one with clinical experience in resuscitation and one with methodological experience. Possible disagreements and discrepancies are resolved by consensus, or by consulting a third reviewer.

We contact the corresponding authors of included studies to obtain key missing data as needed, or to clarify any issues and questions on a particular trial. If authors do not response within two weeks, we contact the corresponding author again, as well as first and senior authors. If this does not lead to an elaboration of the issue, we exclude any outcomes from further analysis whose context cannot be clarified. We only include results into our meta-analysis if their origin is transparent and accurate.

Data items

Data is extracted using a pre-defined data collection form according to the Cochrane Collaboration recommendations (table 5). [65]

Table 5: Data extraction.

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Domain	Data items
Unambiguous study identification	• authors (up to three), year, title, journal or other source, volume, page(s) • identification as conference abstract, or research letter, if applicable • identification number within a study registry, if applicable
Study design	• study type • chronological dimension (prospective, retrospective) • number of study centres (single-centre for 1; multi-centre for ≥5 centres) • characteristics of patients included • type of intervention(s) • case number in total, distributed among intervention and control groups, and subgroups if applicable • mechanical CPR system(s) examined • Guidelines version the study is based on
Results	• qualitative summary • quantitative summary
Risks of bias	• risks of bias according to the Cochrane tool • funding sources • conflicts of interest

Legend to table 5: Data collected from included studies.

Assumptions and simplifications

Cardiac arrest is defined as the absence of cardiovascular circulation. This condition may be validated by evaluating signs of life, palpation of pulse over a major artery, or invasive blood pressure measurement.

Some authors indicate piston-driven mechanical chest compressions as “manual CPR”, particularly when comparing them to compressions by a load-distributing band. Regardless of the denomination chosen by the authors, we only consider chest compressions delivered by hand as “manual CPR”.

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ROSC may be defined differently between studies with regard to the time of its evaluation (at any time, at specific timepoints, or at hospital handover). According to the Ustein definition cited above, we only consider reportings of "ROSC at any time" as our primary outcome "ROSC". Reportings of "ROSC at hospital handover" are considered as secondary outcome "alive at hospital admission".

If the underlying guideline is not stated within the text, we collate the trial to the recommendations valid as per study period.

In some publications, "hands-off time" (in seconds) or "hands-off ratio" (in percentage of "hands-on" or compression time) may be indicated in lieu of "no-flow time" (in seconds).

Risk of bias in individual studies

For each included RCT, we follow Cochrane Risk of Bias tool, which is the standard tool for assessing risks of bias in RCTs. [63] Each of the following potential sources of bias is evaluated where applicable:

- random sequence generation
- allocation concealment
- blinding of participants and personnel
- blinding of outcome assessment
- incomplete outcome data
- selective outcome reporting
- other bias (including baseline imbalance, early termination of the trial, funding bias, etc.)

Each is graded as "high", "low", or "unclear", and a justification for the judgment is presented in the risk of bias table.

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We assess the risk of bias for included quasi-RCT with Cochrane ROBINS-I tool. [66] The following categories are evaluated at study level:

- confounding
- selection of participants into the study
- classification of interventions
- deviations from intended interventions
- missing data
- measurement of outcomes
- selection of the reported result

Each domain is categorized "critical", "serious", "moderate", "low" risk of bias, or "no information". Again, we present justification of the judgement in the risk of bias table. The overall risk of bias for one outcome of quasi-RCT is determined by the highest risk of bias rating of the particular item.

Data synthesis and analysis

Statistical analytics

For dichotomous outcomes (primary and secondary outcomes), odds ratio (OR) with its 95% confidence interval (CI) is used as effect measure per trial. In binary outcomes where rare or 0 events are expected in some trial-arms (i.e. some of the secondary outcomes as trauma), the Peto OR is used to avoid the necessity of a continuity correction. As the Peto OR is only suitable for rare events [63], conventional OR are used for all other binary outcomes. If possible, the OR and its standard error is extracted directly, preferably from an adjusted model. If it is not reported, but the events and numbers of patients per intervention group are available, then the unadjusted OR along with its standard error is obtained. For continuous outcomes, the standardized mean difference (SMD) with its 95% CI is used as effect measure per trial. For

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survival outcomes, (adjusted) HR estimated in Cox regression models are used if reported. Results extracted from Log-rank tests, Kaplan-Meier curves or reported Kaplan-Meier estimates are used alternatively. If neither Cox-regression nor Kaplan-Meier estimators are reported, survival outcomes are extracted as dichotomous outcome. ORs are combined using the Mantel-Haenszel method, and SMDs are pooled by the inverse-variance method. Effect measures are combined separated by study design (RCT and quasi-RCT). [63] Random-effects models are used to account for the expected between-trial heterogeneity in estimating the overall effect and the Hartung-Knapp method is applied for the corresponding CI, if at least 5 trials contribute information to an estimate. [67] Results of fixed-effect models are provided and discussed in contrast to the results of the random-effects models. We investigate all results for statistical heterogeneity by I^2-statistics and by interpreting it in relation to the estimated between-trial variance τ^2. If there is a considerable heterogeneity ($I^2 \geq 75\%$), no meta-analysis is performed.

R version 3.5.0 or higher (R Foundation for Statistical Computing, Vienna, Austria) and the R meta package version 4.9-5 or higher (developed by Guido Schwarzer) are used for all statistical analyses. For generating diagrams, we utilize Review Manager®, Version 5.2.7 (The Nordic Cochrane Centre, Copenhagen, Denmark).

Meta-bias

We perform statistical investigation of potential publication bias based on a test of funnel plot asymmetry, if there are at least 10 studies pooled for one outcome. Sensitivity analyses are conducted with regard to the studies' quality. Further sensitivity analyses are performed during the course of the review, if irregularities of individual studies are found. We set sup "Summary of findings" as recommended by Cochrane Collaboration. [63 65]

In order to detect reporting bias, we compare the data reported in publications with a priori study protocols, or registrations.

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Confidence in cumulative evidence

Quality of evidence is analysed with GRADEpro® GDT (Guideline Development Tool, Grade Working Group, McMaster University and Evidence Prime Inc., Hamilton, Canada). GRADE is an accepted tool for quality assessment and is used to grade the body of evidence at outcome level. [68]

Interventions

Titles and/or abstracts of studies retrieved using the search strategy and those from additional sources after tracing reference sections are screened to identify studies that potentially meet the inclusion criteria. The full text of those potentially eligible studies is retrieved and assessed for eligibility. We use a standardised questionnaire to extract data from the included studies for assessment of study quality and evidence synthesis.

Qualitative synthesis

We provide a narrative synthesis of the findings from the studies included, structured around the mechanical chest compression technique, and summarize study characteristics, including risks of bias and outcomes, in tables.

Quantitative synthesis

Whenever possible, we provide summaries of the intervention effects by using the effect measures along with their 95% CI and p-values for each trial and for the combined effect. The results are graphically illustrated by forest plots.

Subgroups

We plan subgroup analyses for the two different techniques of mechanical chest compression (semi-circumferentially constricting load-distributing band vs. vertically compressing mechanical piston); for the setting (in- vs. out-of-hospital, urban vs. rural, and Europe vs. United States of America vs. other regions); as well as for the initially presented rhythm

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(shockable vs. non-shockable). If applicable, we perform subgroup analysis of aetiology, and concomitant treatment.

Heterogeneity

All results are investigated for statistical heterogeneity by I^2-statistics and by the between-trial variance τ^2. [69] If there is a considerable heterogeneity ($I^2 \geq 75\%$) for an outcome, no meta-analysis will be performed.

Sensitivity analysis is performed to identify scenarios (like non-compliance, protocol violations, missing data, outcome definitions, clustering or correlation, competing risks, baseline imblalance, distibrutional assumptions, or outlies) that influence the findings of the review. [70] Results of sensitivity analysis are presented with the study assessment sheets.

Amendments

Any protocol amendment will be stated with reason within the subsequent publications' "methods" section.

Patient and public involvement

Patients and the publilc were not involved in this protocol of a systematic review and meta-analysis. By defining survival with neurologically intact outcome as primary outcome, we put our focus on the outcome which is most important to the patient. This topic is of specific interest of our patients, because the quality of CPR in general and the quality of chest compressions in particualr will determine their abilities or disabilities, and their quality of life after the incident (i. e. the "neurological outcome"). Unfortunately, this constitutes an endpoint which is not easy to assess and therefore is uncommon in resuscitation studies.

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ETHICS AND DISSEMINATION

No further ethical approval is needed because data from previous studies is retrieved and analysed, whose authors declared clearance by the responsible ethic commissions.

As given by their nature, resuscitation studies cannot meet one crucial issue of the Declaration of HELSINKI Ethical Principles for Medical Research Involving Human Subjects [71] and are regularly conducted under an emergency exception for informed consent. [72-75] Particularly with regard to this predicament, implementation of ethical considerations in resuscitation studies has to be assessed critically.

The publication contains data deriving from a preliminary project as a part of a doctoral thesis at Ulm University. [22 23]

Upon completion, we intend to prepare a manuscript for publication in an international peer-reviewed journal and present the results at scientific conferences. Our scheduled timeline is detailed in table 6.

Table 6: Timeline.

Task / Duration [months]	1	2	3	4	5	6	7	8	9	10
study design and drafting the manuscript for the study protocol	■	■								
critical revision from collaborators		■								
publication phase (study protocol), possibly revisions/re-submissions			■							
study assessment (reviewer #1) study assessment (reviewer #2)				■						
revision of judgements, forging consensus for discrepancies					■					
data extraction (reviewer #1)						■				

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data extraction (reviewer #2)												
qualitative and quantitative synthesis (meta-analyses)												
drafting the manuscripts for publication and presentations												
critical revision from collaborators												
publication phase (meta-analysis), possibly revisions/re-submissions												

Legend to table 6: Scheduled duration of the systematic review and meta-analysis in months.

DISCUSSION

A primary evaluation of this work identified 107 trials on the comparison of mechanical and manual CPR on 179,505 patients, 444 pigs, 62 dogs, and 723 manikin scenarios. This evaluation did not meet established quality criteria of a systematic review, but we were able to become aware of the issues relied with mechanical chest compressions, and current evidence. [22 23]

Several systematic reviews and meta-analyses of heterogeneous design and quality compare outcome rates following manual and mechanical CPR, with their limitations previously mentioned. [22 31 32 34 35 38-44] A carefully performed Bayesian network meta-analysis by Khan and collegues compares manual different mechanical chest compression techniques but misses the opportunity to get a deeper insight into specific subgroups. [76] Traditionally, resuscitation studies assess effects of interventions in overall patient collectives on survival rates in order to answer the question if an intervention is superior than its contol if given to all patients. However, this assumes a similar effect of the same intervention on heterogeneous patients and under varying circumstances. Up to now, only a few findings contribute to solving the clinical problem: if an individual patient may profit from mechanical chest

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compressions and the team should strive for the application an automated mechanical CPR device in a specific occasion. This implies the need of a review which may close this gap in knowledge by putting its focus on the identification of carefully selected subroups concerning patient characertistics, settings, situations, environments, and curcumstances when automated mechanical chest compressions are superior to manual CPR.

Mechanical CPR devices are considered potential advanced life support adjuvants in a public draft of the upcoming guidelines on CPR. While guidelines do not support the devices' routine use, they state several special circumstances when mechanical chest compressions may be considered (e. g. quality, safety, feasibility and practicability issuees). But the recommendations leave it unclear, for which patients, in which situations, or at which point of time mechanical CPR should be aspired. [77]

Strenghts and Limitations

When comparing mechanical to manual CPR, several confounders may influence the patients' outcome, which in their entirety are complex to detect. Any prospective study may be subject to the Hawthorne Effect, when participants show a different behaviour due to the knowledge that they participate in a study. The first automated CPR systems have been developed just shortly after the description of manual chest compression in the 1960s. [78] Therefore, studies over half of a century have to be assessed, and guideline revisions have to be considered in the meantime. On the other hand, it is yet to determine to what extent study designs comply with particular recommendations.

Publication bias is a common problem in evidence-based medicine, as only studies may be assessed if the authors are aware of their existence. In order to reduce publication bias, we ask

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shareholders for unpublished data, but risk of publication bias will depend from their cooperativeness.

Conference abstracts typically are subject to word limits, impairing the possibility to respond to all aspects of study design and conduction in a short text. To prevent bias caused by lacking information from abstracts, we ask the authors for additional information. Again, prevention of bias requires their cooperativeness.

The reliability of this review may be affected by heterogeneous design and quality of the studies included. With our previous work, we became aware of a variability in individual patient characteristics, treatment performance, timing of the devices' installation, and outcome definitions in studies on automated mechanical CPR.

By including quasi-RCT, risks of biases in random sequence generation, blinding, and allocation concealment must be considered. Allocation concealment is generally difficult to assign with an intervention like mechanical chest compression, even in randomized studies. Rescuer blinding is impossible due to the intervention, but patients may be considered blinded because they are unconscious. Though, with predictable risks, we decided to include quasi-RCT into our review.

Surrogate outcomes like ROSC or survival rate until hospital admission are easier to obtain than "hard" clinical outcomes which require considerable resources for study design, funding, recruitment, conduction, and follow-up. But patients are neither interested in survival rates nor in rates of ROSC achieved – the relevant outcome from our patients' point of view is at which neurological state they survive. Hence, neurologically intact survival should be considered of most particular interest.

The most important strength of this systematic review and meta-analysis are the subgroup analyses for automated mechanical chest compression techniques, setting, region, initially

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presented rhythm, aetiology, and concomitant treatment. This is of clinical relevance with regard to increasing treatment options in hospital, where automated mechanical CPR devices might provide a "bridge" to definitive treatment for designated patient groups under certain conditions and in specific environments. We hope that this systematic review and meta-analysis may contribute to implicate future clinical and scientific issues towards an individualized decision-making.

DECLARATIONS

Author Contributions

Study conception and design: MO, JBZ, CMM, KJ, SES

Funding application: MO, KJ, SES, MAW

Acquisition of data: not applicable for study protocols

Data analysis and interpretation: not applicable for study protocols

Drafting of the manuscript: MO, SES

Critical revision for important intellectual content: JBZ, EP, MAW, SW, ADL, CMM, BLN, JTG, SS

All authors read and approved the final manuscript. They agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. MO will have full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

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Competing interests

The authors declare that they have no competing interests.

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Identification	#1a	Identify the report as a protocol of a systematic review	title page
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Amendments			

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	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	n/a; 19
Support			
Sources	#5a	Indicate sources of financial or other support for the review	24
Sponsor	#5b	Provide name for the review funder and / or sponsor	24
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	n/a
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is already known	1-2
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	5-8
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	9-10
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	10-11
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	11-12
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	12-13
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	13
Study records - data collection process	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for	13

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		obtaining and confirming data from investigators	
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	13-14
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	6-8
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	15-16
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	18
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	16-18
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	18-19
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	n/a
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	17
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	17-18

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Automated mechanical cardiopulmonary resuscitation devices versus manual chest compressions in the treatment of cardiac arrest: Protocol of a systematic review and meta-analysis comparing machine to human

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Automated mechanical cardiopulmonary resuscitation devices versus manual chest compressions in the treatment of cardiac arrest:

Protocol of a systematic review and meta-analysis comparing machine to human

TITLE PAGE

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ABSTRACT

Introduction:

Cardiac arrest is a leading cause of death in industrialized countries. Cardiopulmonary Resuscitation (CPR) guidelines follow the principles of closed chest compression as described for the first time in 1960. Mechanical CPR devices are designed to improve chest compression quality, thus considering the improvement of resuscitation outcomes.

This protocol outlines a systematic review and meta-analysis methodology to assess trials investigating the therapeutic effect of automated mechanical CPR devices at the rate of Return of Spontaneous Circulation, neurological state and secondary endpoints (including short- and long-term survival, injuries, and surrogate parameters for CPR quality) in comparison with manual chest compressions in adults with cardiac arrest.

Methods and Analysis:

A sensitive search strategy will be employed in established bibliographic databases from inception until the date of search, followed by forward and backward reference searching. We will include randomized and quasi-randomized trials in qualitative analysis thus comparing mechanical to manual CPR. Studies reporting survival outcomes will be included in quantitative analysis.

Two reviewers will assess independently publications using a pre-defined data collection form. Standardized tools will be used for data extraction, risks of bias and quality of evidence. If enough studies are identified for meta-analysis, the measures of association will be calculated by dint of bivariate random-effects models. Statistical heterogeneity will be evaluated by I^2-statistics and explored through sensitivity analysis. By comprehensive

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subgroup analysis we intend to identify subpopulations who may benefit from mechanical or manual CPR techniques. The reporting follows the PRISMA statement.

Ethics and dissemination:

No ethical approval will be needed because data from previous studies will be retrieved and analysed. Most resuscitation studies are conducted under an emergency exception for informed consent. This publication contains data deriving from a dissertation project. We will disseminate the results through publication in a peer-reviewed journal and at scientific conferences.

Registration:

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ARTICLE SUMMARY

Strengths and limitations of this study

- This systematic review and meta-analysis addresses gaps in the current evidence-base on chest compression techniques as part of cardiopulmonary resuscitation by providing a comprehensive assessment of randomized and quasi-randomized clinical trials, and comprehensive subgroup analysis.
- Extensive and differentiated multi-step search strategy including 12 databases without language restrictions.
- Methods follow the Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) Protocols guidelines, Meta-Analysis Of Observational Studies in Epidemiology (MOOSE) Statement, and Cochrane Handbook.
- This review is limited to randomized and quasi-randomized clinical trials of heterogeneous design and quality.
- Potential clarification which subpopulations, in which situations, under which circumstances, and at which time may benefit from which chest compression technique, may improve treatment decisions.

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INTRODUCTION

For thousands of years, mankind has struggled to "re-animate" people showing no signs of life. A milestone of Cardiopulmonary Resuscitation (CPR), manual closed-chest cardiac massage, was introduced in 1960. [1] More recent developments are machines that perform chest compressions mechanically and automatically through inflatable vests, pistons, or load distributing bands. But can automated mechanical chest compression devices surpass human performance, and again revolutionize CPR?

Rationale

Mechanical CPR may represent a method of improving compression quality in certain clinical scenarios and for chosen subpopulations. Many patients who suffer sudden cardiac arrest have the potential for recovery with good quality of life, but this recovery hinges on high quality CPR at the time of the event. [2-4]

Burden of disease

Sudden cardiac arrest following coronary heart disease is a major cause of death in industrialized nations with an age-related prevalence of 80-150 per 100,000 person years and mortality rates as high as 80-90%. In about 10% of patients, sudden cardiac arrest following an acute myocardial infarction is the first manifestation of a (previously unknown) coronary heart disease. In younger patients, ion channel disease, hereditary cardiomyopathies, and myocarditis are frequent causes of sudden cardiac arrest. [2-9]

Reflecting this special group, sudden cardiac arrest is a vast socio-economic burden, as these patients are at the height of their careers. With the progress of modern medicine, even elderly or handicapped people have the potential to recover from an incident with a good quality of life. Beyond the vast social burden of losing lives, to the pure healthcare-related costs which are estimated $ 33 billion per year in the United States, [10] losses of earnings and productivity

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have to be added in order to realize the costs across the entire society. Though, the burden of premature death in years of potential life lost is greater for sudden cardiac arrest than for all individual cancers and most other leading causes of death. [11]

What is known?

The medical problem

CPR quality is important for resuscitation success. With manual CPR, increasing fatigue of the rescuers and frequent interruptions of compressions have been reported. Both fatigue and interruptions decrease blood flow required for adequate myocardial and brain perfusion which is crucial for ROSC and good neurological outcome. Compared to insufficiently performed chest compressions, chest compressions of good quality and with little interruptions have been shown to improve survival rates. [12-14]

The issues encountered by manual CPR led to the development of mechanical CPR systems which are supposed to improve CPR quality, an essential determinant of survival. [14 15] Application studies have shown that by using mechanical CPR devices higher arterial carbon dioxide levels, which are an indicative of better tissue perfusion, can be achieved [16-18] as well as better haemodynamic conditions, thereby leading to improved coronary and cerebral blood flow. [16 19-21] Furthermore, it has been stated that coronary and brain perfusion are the leading determinants of survival following resuscitation. Based on the premise that automated mechanical CPR provides a sustained quality of chest compressions, better outcomes might be expected from these devices. [22]

A search in literature, carried out in a primary evaluation as a supplemental appraisal of a medical thesis, showed numerous studies of heterogeneous design, many of poor quality. [22 23] Three high-quality, prospective, large, multi-centre randomized controlled trials on mechanical CPR (CIRC, [24] LINC [25] and PARAMEDIC [26]) have been published, as well as

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several publications from retrospective registry analyses. [22 27-30] They find inconsistent and contradictory results, leaving it unclear if manual or mechanical chest compressions during CPR may improve the outcome of patients suffering from cardiac arrest. Considered more in detail, we neither have evidence if there is a special point of time to install a mechanical CPR device, nor if there are special situations or subgroups of patients who will particularly benefit from mechanical chest compressions. In particular, we do not know if one compression technique (vertical piston versus semi-circumferential load-distributing band) is superior. [22 31]

Need for the systematic review

A recently published Cochrane Review shows several weaknesses: [31] Most important, the intended subgroup analysis to distinguish vertical chest compressions by a piston from semi-circumferentially constricting compressions by a load-distributing band, and in- from out-of-hospital cardiac arrest, was abandoned due to the variability of included trials. Therefore, this meta-analysis does not consider different treatments, which may have different or even contrary effects on the outcomes, which might increase between-trial heterogeneity unnecessarily.

Beyond the Cochrane review, several other meta-analyses of poor quality have been published. One review was sponsored by the manufacturer of a mechanical CPR device. [32] As this study shows a superiority of the device produced by the sponsor of the review to the competitors' device, and to manual CPR, the reader should be aware of conflicts of interest by the authors affiliated with this company. This has been criticized, as "industry sponsored meta-analyses are more likely to recommend an intervention". [33] Moreover, this particular study examined ROSC only, but no long-term outcomes, and included both clinical and animal models. Possible conflicts of interest may also be considered if authors conducting the systematic review co-authored a trial which is included in the review, as they have to assess critically their own publications. [24 26 34 35]

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Another review provides an overview of existing systematic reviews and studies and comes to the conclusion that there are more studies needed for a definitive judgement. [36] Systematic reviews are at increased risk of selection and attrition bias and confounding when observational studies are included, [37] which applies to some reviews. [22 32 35 38] Further reviews do not distinguish between different chest compression techniques in their meta-analysis, as described above, [31 35 38 39] or only include one particular device. [40-42] Respectively, others focus only on in- [43] or out-of-hospital [32 34 35 39 41 42] cardiac arrest.

Some reviewers only search Medline®, the most important database, though under-representing publications indexed to other databases. As numerous European journals are indexed to EMBASE®, this may have led to selection bias and an over-representation of American trials (or trials published in American Journals). As two of the main competitors have been developed in the US and in Europe, this search strategy may not be representative. Furthermore, we have to consider that there are fundamental differences between the American paramedic-based emergency medical service system and the physician-staffed systems as largely provided in European countries. [32 36 44]

Adverse effects

Important adverse effects of mechanical CPR systems are injuries caused by chest compression. They comprise bone fractures (e.g., ribs, sternum, vertebrae) and organ lacerations (e.g., liver, spleen). [45] Injuries do not seem to be more frequent than those caused by manual chest compressions, but the pattern may differ. [22 46 47] The installation of the device may delay the further CPR process and therefore delay life-saving measures like defibrillation and supporting measures like airway management or intravenous access. [27 48-51] And last but not least, as it goes for any technical device, failures may occur.

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Objectives

Patients

The objective of this systematic review and meta-analysis is to detect whether adults suffering from cardiac arrest, or specific subpopulations (setting urban vs. rural, region, in-hospital vs. out-of-hospital, and if applicable initially presented rhythm, aetiology, and concomitant treatment) may benefit from the application of automated mechanical CPR devices.

Experimental intervention

Mechanical CPR devices deliver chest compressions independently of the user, either through vertical force by a piston, or through semi-circumferentially constricting force by a load-distributing band. So far, it is unknown if one compression technique might be superior. Mechanical CPR devices are also referred to as “automated CPR systems”, or “external cardiac compressors”.

Until the application of the mechanical CPR device, the patients may have received manual chest compressions as described below. To date, we still do not know if the timing of the application of the devices is of importance.

Control intervention

Conventional manual chest compressions follow valid recommendations. These guidelines are mandatory worldwide and they summarize all the evidence for treatment recommendations and describe the best clinical practice. Furthermore, manual chest compressions are generally available. Hence, manual CPR only is considered as the comparator.

Minimum requirements for conventional manual chest compressions include 100-120 vertical compressions per minute of approximately one third of the sternum height, performed at the middle of chest (caudal third of sternum, intermammary line) in supine position. [52]

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Outcomes and prioritization

ROSC is one of our primary outcomes. This common endpoint in resuscitation studies can be easily obtained and does not depend on a clinical follow-up. ROSC, which is measured as binary variable, is defined as any spontaneous pulse "detectable by manual palpation of a major artery" at any time following resuscitation attempts. [53]

The other primary outcome is the neurologic state according to the Glasgow-Pittsburgh cerebral performance category (CPC) [54], or modified Rankin score (mRS) [55]; the association between CPC and mRS is described elsewhere [56]. The neurological outcome is evaluated after 30 days, or at hospital discharge. If the CPC scores of 1 and 2 are only reported in a summarized form as an "intact" or "good" neurological outcome, they are treated as a binary event ("good" vs. "poor"). [54] By way of derogation from our initial study registration, we consider neurologically intact survival as a primary outcome because of the essential importance of this outcome for the individual patients' further lives.

Secondary outcomes include short- and long-term survival, resuscitation trauma, and CPR quality surrogates (table 1).

Table 1: Secondary outcomes.

Domain	Secondary outcome
Short-term survival	• alive at hospital admission • alive ≤30 days after incident • alive at hospital discharge
Long-term survival	• alive 3, 6, and 12 months after incident • health-related quality of life [57] at each time point • neurologic state [54-56] at each time point
Resuscitation trauma	• frequency of injuries and trauma caused by chest compressions (including skin lesions, skeletal fractures, pulmonary lesions,

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	pneumo-, haemothorax, haemoperitoneum, organ lacerations)
CPR quality surrogates	• guideline compliance (including chest compression depth, frequency, rate, ratio) • no-flow (hands-off) time, and ratio • arterial blood pressure during chest compressions • end-tidal carbon dioxide partial pressure ($etCO_2$) with ongoing CPR

Legend to table 1: CPR = cardiopulmonary resuscitation; $etCO_2$ = end-tidal carbon dioxide.

CPR quality surrogates may have an impact on patient survival, as fatigue and interruptions decrease the blood flow required for adequate myocardial and brain perfusion, which is crucial for a favourable neurological outcome. [58 59] Compared to insufficiently performed chest compressions, high fractions of chest compressions have been shown to be associated with improved survival rates. [14 60-62] Arterial blood pressure is used to describe hemodynamics, as it determines cerebral and coronary perfusion. End-tidal carbon dioxide ($etCO_2$) is considered a surrogate marker for induced cardiac output.

Primary and secondary outcomes are reported according to the "Utstein Style", a protocol for uniform reporting of resuscitation outcomes, [53] and the "Core Outcome Set for Cardiac Arrest (COSCA)" [57], based in parts of previous reviews.

Review questions

The review of studies is designed to address the following questions:

- Is a specific CPR technique, manual or mechanical (by band or piston as subgroups) associated with an improved primary or secondary outcome, as stated above?
- May automated mechanical CPR devices improve the process, quality, and safety of CPR?

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- What are possible scopes of application for automated mechanical CPR devices? This question seeks to ascertain which patient subgroups in which situations, under which circumstances, and at which time may benefit from automated mechanical CPR.

METHODS AND ANALYSIS

We registered this study at the International Prospective Register of Systematic Reviews (PROSPERO), registration number CRD42017051633. The reporting of this protocol accords to the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) statement. [63] The PRISMA-P checklist is provided as an additional file. We will follow both Cochrane Handbook [64] and MOOSE Statement [65].

Eligibility criteria

Study characteristics

Eligible for inclusion will be randomized controlled trials (RCT) and quasi-RCT comparing mechanical to manual chest compressions in cardiac arrest if at least one primary endpoint or other survival outcome is available. In this context, state (published, accepted for publication, or pre-print) and mode of publication (full article or conference abstract) will be irrelevant. Studies will only be eligible for inclusion if the data for the control group is published.

Population characteristics and setting

We will only include studies in adult patients (≥18 years of age), as most mechanical CPR devices are not approved for children. We will consider events as "in-hospital" if the cardiac arrest occurs to a hospitalized patient and if the Medical Emergency Team of the institution is activated. Events treated by official Emergency Medical Services outside hospital buildings will be considered as "out-of-hospital". Both settings, in- and out-of-hospital cardiac arrest, will be eligible for inclusion.

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Device characteristics

All mechanical CPR devices that deliver automated chest compressions will be included in the systematic review to gain comprehensive insight into the topic and to avoid selection bias. We identified available devices by a preliminary study [22 23] and subsequent search via the internet. The devices may compress the chest independently of the user, either through vertical force (applied by a piston) or semi-circumferentially constricting force (applied by a load-distributing band). However, we will conduct subgroup analysis for each compression technique. Technical characteristics differ for each device (table 2). [22]

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Table 2: Specifications of automated mechanical CPR devices.

device	AutoPulse®	LUCAS® 3	LifeStat® 1008	Corpuls® CPR	LifeLine® ARM	Easy Pulse®	MCC® Mini Chest Compressor	X-CPR® 2
manufacturer	Zoll Medical, Chelmsford, USA (prior: Revivant)	Stryker, Kalamazoo, USA (prior: Physio-Control, Jolife)	Michigan Instruments, Grand Rapids, USA	GS Elektromedizinische Geräte G. Stemple, Kaufering, Germany	Defibtech, Guilford, USA	Schiller, Baar, Switzerland	Resuscitation International, Scottsdale, USA	CU Medical Systems, Wonju, Korea (prior: Healthwell Medical)
technique	semi-circumferentially constricting load distributing band, electric driven	piston, electric driven	piston & ventilator, pneumatic driven	piston, electric driven	piston, electric driven	piston, electric driven, non-constricting band	Piston, pneumatic driven, non-constricting band	piston with semi-circumferentially constricting thoracic strap, electric driven
compression - depth	20% of thorax diameter + 0.6 / - 1.3 cm	5.3 ± 0,2 cm (4.0–5.3 cm for sternum height <18.5 cm)	0.0–8.0 cm (adjustable)	2.0–6.0 cm (adjustable)	5.3 ± 0.3 cm	4.0–5.2 cm (adjusted for chest circumference or width)	3.8 or 5.1 cm (selectable)	[no information provided]
- rate	80 ± 5 /min	102 ± 2 /min	100 ± 6 /min	80 – 120 /min (adjustable)	101 ± 1 /min	100 /min	110 ± 22 /min	[no information provided]
compression / ventilation ratio (c/v)	30:2, 15:2, or continuous compression (selectable)	30:2 or continuous compression (selectable)	30:2 or continuous compression (selectable)	30:2, 15:2, or continuous compression (selectable)	30:2 or continuous compression (selectable), updates possible	30:2 or continuous compression (selectable)	continuous compression	[no information provided]
patient specifications	chest circumference 76–130 cm chest width 25–38 cm weight ≤136 kg	sternum height 17–30 cm chest width ≤45 cm	sternum height 11–37 cm chest width ≤56 cm	sternum height 14–34 cm chest width ≤48 cm	sternum height 17–30 cm chest width ≤46 cm	chest circumference 76–135 cm chest width 22–40 cm	chest circumference 78 – 130 cm	[no information provided]
device dimensions - size	83 × 45 × 8 cm + load-distributing band & battery	58 × 33 × 26 cm (stowed) 56 × 52 × 24 cm (assembled)	57 × 46 × 19 cm (assembled)	45 × 43 × 9 cm + board (different sizes)	51 × 51 × 25 cm (stowed) 60 × 53 × 23 cm (assembled)	18 × 22 × 15 cm	20 × 16 × 15 cm + band	[no information provided]
- weight	11.6 kg including battery	8.0 kg including battery + straps	8.9 kg + gas supply	5.5 kg including battery + board (1.7–2.1 kg)	7.1 kg including battery	3.5 kg + belt with slider (0.3 kg)	2.0 kg + gas supply	[no information provided]
- average consumption	battery run-time ≈ 30 min	battery run-time ≈ 45 min external power supply	≤45 l/min compressed gas	battery run-time ≈ 90 min, rechargeable during use	battery run-time ≈ 60 min	battery run-time ≈ 45 min external power supply	60 ± 5 l/min compressed gas	[no information provided]

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features	event recorder	report generator	n/a	memory card interface	n/a	computer interface	n/a	n/a
forerunner models	**vest CPR**: inflatable vest, pneumatic driven, circumferential pressure 200-280 mmHg, compression rate 60 /min	**LUCAS®**: pneumatic driven piston, continuous compressions **LUCAS®** 2: electric driven piston	**thumper®** (under licence for Europe: Siemens **Sirepuls®**): model *1001*, *1002*, *1007CC*: no ventilator, continuous compressions *1003*, *1004*, *1005*, *1007*, *1007CCV*: C/V 5:1 or continuous compressions	n/a	n/a	n/a	n/a currently not selling in Europe & USA	**X-CPR®** (SST-CPR®), pneumatic driven piston & strap currently not selling

Legend to table 2: Specifications according to manufacturer information. Abbreviations: CPR = cardiopulmonary resuscitation; c/v = compression/ventilation ratio; n/a = not applicable; mmHg = millimetres of mercury.

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Before the first application of the automated CPR device, patients may receive manual chest compressions as described above.

Exclusion criteria

Case-series, case-control-studies, or historically controlled studies will be excluded, as well as studies involving traumatic cardiac arrest, as treatment priorities fundamentally differ from non-traumatic entities. We expect this study population to represent patients suffering from sudden cardiac arrest, which is predominantly a cardiac aetiology.

We will not apply any restrictions on manufacturer, model, technique, or publication language.

Information sources

We will consider relevant bibliographic databases, as listed in table 3, from inception until the date of search and document the date of our last access on the search engine.

Table 3: Searched databases.

database
Journals@Ovid Full Text
Your Journals@Ovid
EBM Reviews - Cochrane Database of Systematic Reviews - ACP Journal Club - Database of Abstracts of Reviews of Effects - Cochrane Central Register of Controlled Trials - Cochrane Methodology Register - Health Technology Assessment - NHS Economic Evaluation Database
BIOSIS Previews®

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Embase®
Ovid MEDLINE® - in-process & other non-indexed citations - daily update

Legend to table 3: EBM = evidence-based medicine; ACP = American College of Physicians; NHS = National Health Service.

Search strategy and study selection

Search strategy

Our search strategy will comprise several steps: We will start with a combined, structured, and keyword-based search containing search terms in the field of cardiopulmonary resuscitation, and the names of devices and techniques, as presented in table 4. Backward reference searching may identify further literature eligible for inclusion by analysing the reference sections of the articles found by our search strategy. Forward reference searching will be executed to identify additional relevant studies that cite a study included. Pertinent study registries will be searched via the World Health Organization (WHO) International Clinical Trials Registry Platform (ICTRP; http://apps.who.int/trialsearch) for trials, which have been registered, but not published. Finally, we will ask manufacturers of mechanical CPR devices for unpublished data, and we will assess preprint servers and data repositories.

Table 4: Keyword-based search.

#	search item
1	mechanical chest compressors
2	mechanical CPR device
3	chest compression device
4	mechanical cardiopulmonary resuscitation
5	vest CPR
6	Thumper

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7	AutoPulse
8	Lund University Cardiac Assist System
9	LUCAS-2 *or* LUCAS2 *or* LUCAS-3 *or* LUCAS3
10	LifeStat *or* Life-Stat
11	Miniaturized Chest Compressor
12	Animax
13	Corpuls CPR
14	LifeLinie ARM
15	EasyPulse
16	X-CPR
17	load distributing band
18	mechanical piston
19	resuscitation/ *or* resuscitator/ *or* advanced life support
20	1 *or* 2 *or* 3 *or* 4 *or* 5 *or* 6 *or* 7 *or* 8 *or* 9 *or* 10 *or* 11 *or* 12 *or* 13 *or* 14 *or* 15 *or* 16 *or* 17 *or* 18
21	19 *and* 20
22	*remove duplicates from* 21

Legend to table 4: Multiple entries are feasible. CPR = cardiopulmonary resuscitation; a slash indicates Medical Subject Headings (MeSH) terms.

For literature search, we will use the proprietary search engine Ovid®, Ovid Technologies Inc., Wolters Kluwer Health, New York/USA.

Study selection

We will start by screening both the title and the abstract of the publications identified through our search for eligibility. If the study meets any inclusion but no exclusion criteria, we will assess the full text in order to extract study characteristics, results, and risks of bias, as stated below.

Through our search strategy (table 4), duplicates will automatically be removed. Nevertheless, additional duplicates may occur if results of one study are published in several manuscripts

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with different titles, in different languages, or different modes of publication. We will regard publications as duplicates if the same authors, or authors from the same centres, publish studies covering the same observation period, with comparable case numbers, similar design, and related study objectives. In case of redundant publication, we will prefer full texts to abstracts, and we will merge the published data.

Study records

Data management

The characteristics and results of the studies included will be extracted and summarized in a table (Numbers® 6.2, Apple Inc., Cupertino/USA). If eligible, frequencies of primary and secondary outcomes will be charted in forest plots (Review Manager® 5.3, The Nordic Cochrane Centre, Copenhagen/Denmark).

Selection process and collection of data

Study selection, quality assessment, and data extraction will be performed independently by two reviewers, one of whom with clinical experience in resuscitation and one with methodological experience. Possible disagreements and discrepancies will be resolved by consensus or by consulting a third reviewer.

We will contact the corresponding authors of included studies to obtain key missing data as needed, or to clarify any issues and questions on a particular trial. If authors do not respond within two weeks, we will contact the corresponding author again, as well as first and senior authors. If this does not lead to an elaboration of the issue, we will exclude any outcomes whose context cannot be clarified from further analysis. We will only include results in our meta-analysis if their origin is transparent and accurate.

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Data items

Data will be extracted by using a pre-defined data collection form according to the Cochrane Collaboration recommendations (table 5). [66]

Table 5: Data extraction.

Domain	Data items
Unambiguous study identification	• authors (up to three), year, title, journal or other source, volume, page(s) • identification as conference abstract, or research letter, if applicable • identification number within a study registry, if applicable
Study design	• study type • chronological dimension (prospective, retrospective) • number of study centres (single-centre for 1; multi-centre for ≥5 centres) • characteristics of patients included • type of intervention(s) • case number in total, distributed among intervention and control groups, and subgroups if applicable • mechanical CPR system(s) examined • guidelines version the study is based on
Results	• qualitative summary • quantitative summary
Risks of bias	• risks of bias according to the Cochrane tool • funding sources • conflicts of interest

Legend to table 5: Data collected from included studies.

Assumptions and simplifications

Cardiac arrest is defined as the absence of cardiovascular circulation. This condition may be validated by evaluating signs of life, palpation of pulse of a major artery, or invasive blood pressure measurement.

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Some authors indicate piston-driven mechanical chest compressions as "manual CPR", particularly when comparing them to compressions by a load-distributing band. Regardless of the denomination chosen by the authors, we will only consider chest compressions delivered by hand as "manual CPR".

ROSC may be defined differently in studies with regard to the time of its evaluation (at any time, at specific timepoints, or at hospital handover). According to the Utstein definition cited above, we will only consider reportings of "ROSC at any time" as our primary outcome "ROSC". Reportings of "ROSC at hospital handover" will be considered as secondary outcome "alive at hospital admission".

If the underlying guideline is not stated within the text, we will collate the trial to the recommendations which were valid as per period of study.

In some publications, "hands-off time" (in seconds) or "hands-off ratio" (in percentage of "hands-on" or compression time) may be indicated in lieu of "no-flow time" (in seconds).

Risk of bias in individual studies

For each included RCT, we will follow Cochrane Risk of Bias tool, which is the standard tool for assessing risks of bias in RCTs. [64] Each of the following potential sources of bias will be evaluated where applicable:

- random sequence generation
- allocation concealment
- blinding of participants and personnel
- blinding of outcome assessment
- incomplete outcome data
- selective outcome reporting

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- other bias (including baseline imbalance, early termination of the trial, funding bias, etc.)

Each will be graded as "high", "low", or "unclear", and a justification for the judgment will be presented in the risk of bias table.

We will assess the risk of bias for included quasi-RCT with Cochrane ROBINS-I tool. [67] The following categories will be evaluated at study level:

- confounding
- selection of participants into the study
- classification of interventions
- deviations from intended interventions
- missing data
- measurement of outcomes
- selection of the reported result

Each domain will be categorized "critical", "serious", "moderate", "low" risk of bias, or "no information". We will also present the justification of the judgement in the risk of bias table. The overall risk of bias for one outcome of quasi-RCT will be determined by the highest risk of bias rating of the particular item.

Data synthesis and analysis

Statistical analytics

For dichotomous outcomes (primary and secondary outcomes), odds ratio (OR) with its 95% confidence interval (CI) will be used as effect measure per trial. In binary outcomes where rare or 0 events are expected in some trial-arms (i.e., some of the secondary outcomes as trauma), the Peto OR will be used to avoid the necessity of a continuity correction. As the

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Peto OR is only suitable for rare events [64], conventional OR will be used for all other binary outcomes. If possible, the OR and its standard error will be extracted directly, preferably from an adjusted model. If the adjusted OR is not reported, the unadjusted OR along with its standard error will be obtained when the events and numbers of patients per intervention group are available. For continuous outcomes, the standardized mean difference (SMD) with its 95% CI will be used as effect measure per trial. For survival outcomes, (adjusted) HR estimated in Cox regression models will be used if reported. Results extracted from Log-rank tests, Kaplan-Meier curves or reported Kaplan-Meier estimates will be used as alternatives. If neither Cox-regression nor Kaplan-Meier estimators are reported, survival outcomes will be extracted as dichotomous outcome. ORs will be combined by using the Mantel-Haenszel method, and SMDs will be pooled by the inverse-variance method. Effect measures will be combined separated by study design (RCT and quasi-RCT). [64] Random-effects models will be used to account for the expected between-trial heterogeneity by estimating the overall effect, and the Hartung-Knapp method will be applied to the corresponding CI if at least 5 trials contribute information to an estimate. [68] Results of fixed-effects models will be provided and discussed in contrast to the results of the random-effects models. We will investigate all results for statistical heterogeneity by I^2-statistics and by interpreting them in relation to the estimated between-trial variance τ^2. If there is a considerable heterogeneity ($I^2 \geq 75\%$), no meta-analysis will be performed.

R version 3.5.0 or higher (R Foundation for Statistical Computing, Vienna, Austria) and the R meta package version 4.9-5 or higher (developed by Guido Schwarzer) will be used for all statistical analyses. For generating diagrams, we will utilize Review Manager®, Version 5.2.7 (The Nordic Cochrane Centre, Copenhagen, Denmark).

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Meta-bias

We will perform a statistical investigation of potential publication bias based on a test of funnel plot asymmetry, if there have been at least 10 studies pooled for one outcome. Sensitivity analyses will be conducted with regard to the quality of studies. Further sensitivity analyses will be performed during the course of the review, if irregularities of individual studies are found. We will set up "summary of findings" as recommended by Cochrane Collaboration. [64 66]

In order to detect reporting bias, we will compare the data reported in publications with a priori study protocols, or registrations.

Confidence in cumulative evidence

The quality of evidence will be analysed by using GRADEpro® GDT (Guideline Development Tool, Grade Working Group, McMaster University and Evidence Prime Inc., Hamilton, Canada). GRADE is an accepted tool for quality assessment and is used to grade the body of evidence at outcome level. [69]

Interventions

Titles and/or abstracts of studies retrieved from the search and those from additional sources after tracing reference sections will be screened to identify studies which possibly meet the inclusion criteria. The full text of those potentially eligible studies will be retrieved and assessed for eligibility. We will use a standardized questionnaire to extract data from the included studies for assessment of study quality and evidence synthesis.

Qualitative synthesis

We will provide a narrative synthesis of the findings from the studies included, which will be structured around the mechanical chest compression techniques, and we will summarize study characteristics by including risks of bias and outcomes, in tables.

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Quantitative synthesis

Whenever possible, we will provide summaries of the intervention effects by using the effect measures along with their 95% CI and p-values for each trial and for the combined effect. The results will be graphically illustrated by forest plots.

Subgroups

We plan subgroup analyses for both techniques of mechanical chest compression (semi-circumferentially constricting load-distributing band vs. vertically compressing piston), for the setting (in- vs. out-of-hospital, urban vs. rural, and Europe vs. United States of America vs. other regions) as well as for the initially presented rhythm (shockable vs. non-shockable). If applicable, we will perform subgroup analysis of aetiology and concomitant treatment.

Heterogeneity

All results will be investigated for statistical heterogeneity by I^2-statistics and by between-trial variance τ^2. [70] If there is a considerable heterogeneity ($I^2 \geq 75\%$) for an outcome, no meta-analysis will be performed.

Sensitivity analysis will be performed to identify scenarios (like non-compliance, protocol violations, missing data, outcome definitions, clustering or correlation, competing risks, baseline imbalance, distributional assumptions, or outliers) which influence the findings of the review. [71] Results of sensitivity analysis will be presented together with the study assessment sheets.

Amendments

Any protocol amendment will be stated with reason in the "methods" section of the subsequent publications.

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Patient and public involvement

Patients and the public have not been involved in this protocol of a systematic review and meta-analysis. By defining neurologically intact survival as primary outcome, we will put our focus on the outcome which is most important to the patient. This topic is of specific interest to our patients, because the quality of CPR in general and the quality of chest compressions in particular will determine their abilities or disabilities, and their quality of life after the incident (i. e. the "neurological outcome"). Unfortunately, this constitutes an endpoint which is not easy to assess and therefore is uncommon in resuscitation studies.

ETHICS AND DISSEMINATION

No further ethical approval is needed as data from previous studies, whose authors declared clearance by the responsible ethics commissions, will be retrieved and analysed.

As resuscitation studies are conducted under an emergency, they cannot meet the demand of informed consent, which is an important issue laid down in the Declaration of HELSINKI Ethical Principles for Medical Research Involving Human Subjects [72] Though, these studies regularly use an emergency exception to the need for informed consent [73-76]. In particular with regard to this predicament, the implementation of ethical considerations in resuscitation studies has to be assessed critically.

This publication contains data deriving from a preliminary project which was part of a doctoral thesis at Ulm University. [22 23]

After the completion of this meta-analysis, we intend to prepare a manuscript for publication in an international peer-reviewed journal and to present the results at scientific conferences. Our scheduled timeline is detailed in table 6.

Table 6: Timeline.

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Task / Duration [months]		1	2	3	4	5	6	7	8	9	10	
study design and drafting the manuscript for the study protocol												
critical revision by collaborators												
publication phase (study protocol), possibly revisions/re-submissions												
study assessment (reviewer #1) study assessment (reviewer #2)												
revision of judgements, forging consensus for discrepancies												
data extraction (reviewer #1) data extraction (reviewer #2)												
qualitative and quantitative synthesis (meta-analyses)												
drafting the manuscripts for publication and presentations												
critical revision by collaborators												
publication phase (meta-analysis), possibly revisions/re-submissions												

Legend to table 6: Scheduled duration of the systematic review and meta-analysis in months.

DISCUSSION

A primary evaluation of this project identified 107 trials on 179,505 patients, 444 pigs, 62 dogs, and 723 manikin scenarios comparing mechanical to manual CPR. This evaluation did not meet established quality criteria of a systematic review, but we were able to become aware of the issues related to mechanical chest compressions and current evidence. [22 23]

Several systematic reviews and meta-analyses of heterogeneous design and quality compare outcome rates following manual and mechanical CPR, with their limitations previously

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mentioned. [22 31 32 34 35 38-44] A carefully performed Bayesian network meta-analysis by Khan and colleagues compares manually operated chest compressions with different mechanical chest compression techniques, but misses the opportunity to get a deeper insight into specific subgroups. [77] Traditionally, resuscitation studies assess effects of interventions in overall patient collectives on survival rates in order to answer the question if an intervention is superior to its control. However, this assumes a similar effect of the same intervention in heterogeneous patients and under varying circumstances. Up to now, only a few findings contribute to solving the clinical problem: if an individual patient may profit from mechanical chest compressions and if the team should strive for the application of an automated mechanical CPR device on a specific occasion. This implies the need of a review which may close this gap in knowledge by putting its focus on the identification of carefully selected subgroups concerning patient characteristics, settings, situations, environments and circumstances when automated mechanical chest compressions are superior to manual CPR.

Mechanical CPR devices are considered potential advanced life support adjuvants in a public draft of the upcoming guidelines on CPR. While guidelines do not support the routine use of these devices, they state several special circumstances when mechanical chest compressions may be considered (issues such as quality, safety, feasibility and practicability). But the recommendations do not make it clear when mechanical CPR should be considered, that is for which patients, in which situations, or at which point of time. [78]

Strengths and limitations

When comparing mechanical to manual CPR, several confounders may influence the patients' outcome, which in their entirety are complex to detect. Any prospective study may be subject to a Hawthorne-like effect, when individuals (in this case, emergency or critical care teams) show a different behaviour due to the knowledge that they participate in a study leading to performance bias in or between intervention and control groups. The first automated CPR

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systems have been developed just shortly after the description of manual chest compression in the 1960s. [79] Therefore, studies of over half a century will have to be assessed, and guideline revisions will have to be considered in the meantime. On the other hand, it is yet to determine to what extent study designs comply with particular recommendations.

Publication bias is a common problem in evidence-based medicine, as only studies may be assessed if the authors are aware of their existence. In order to reduce publication bias, we will ask stakeholders for unpublished data, but the risk of publication bias will depend on their cooperativeness.

Conference abstracts are typically subject to word limits, impairing the possibility of responding to all aspects of study design and conduction in a short text. To prevent bias due to the lack of information in abstracts, we will ask the authors for additional information. As mentioned above, the prevention of bias will require their cooperativeness.

The reliability of this review may be affected by heterogeneous design and quality of the studies included. Through our previous work, we have become aware of a variability in individual patient characteristics, treatment performance, timing of the installation of devices, and outcome definitions in studies of automated mechanical CPR.

By including quasi-RCT, risks of bias in random sequence generation, blinding, and allocation concealment must be considered. It is generally difficult to assign allocation concealment to an intervention like mechanical chest compression, even in randomized studies. The blinding of rescuers is impossible due to the intervention, but patients may be considered blinded because they are unconscious. Though, with predictable risks, we decided to include quasi-RCT in our review.

Surrogate outcomes like ROSC or survival rate until hospital admission are easier to obtain than “hard” clinical outcomes which require considerable resources for study design, funding,

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recruitment, conduction and follow-up. But patients are interested neither in survival rates nor in rates of ROSC – the relevant outcome from our patients' point of view is at which neurological state they may survive. Hence, neurologically intact survival should be considered as a matter of most particular interest.

The most important strength of this systematic review and meta-analysis will be the subgroup analyses of automated mechanical chest compression techniques, setting, region, initially presented rhythm, aetiology, and concomitant treatment. This is of clinical relevance with regard to increasing treatment options in hospital, where automated mechanical CPR devices might provide a "bridge" to definitive treatment for designated patient groups under certain conditions and in specific environments. We hope that this systematic review and meta-analysis may contribute to implicate future clinical and scientific issues towards an individualized decision-making.

DECLARATIONS

Author Contributions

Study conception and design: MO, JBZ, CMM, KJ, SES

Funding application: MO, KJ, SES, MAW

Acquisition of data: not applicable to study protocols

Data analysis and interpretation: not applicable to study protocols

Drafting of the manuscript: MO, SES

Critical revision for important intellectual content: JBZ, EP, MAW, SW, ADL, CMM, BLN, JTG, SS

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All authors read and approved of the final manuscript. They agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. MO will have full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

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Competing interests

The authors declare that they have no competing interests.

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Reporting checklist for protocol of a systematic review.

Based on the PRISMA-P guidelines.

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		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	title page
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	n/a
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	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	13
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Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	31-32
Amendments			
	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	n/a; 26
Support			
Sources	#5a	Indicate sources of financial or other support for the review	32
Sponsor	#5b	Provide name for the review funder and / or sponsor	32
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	n/a
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is already known	6
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	9-13
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	13-17
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	17-18
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	18-20
Study records -	#11a	Describe the mechanism(s) that will be used to manage	20-21

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data management		records and data throughout the review	
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	20
Study records - data collection process	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	20
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	21
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	11-12
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	22-23
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	25
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	23-26
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	26
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	n/a
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	25

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Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	25

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